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Wang et al.

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(54) **NON-VOLATILE MEMORY WITH
OVERDRIVE VOLTAGE ZONING TO
COMPENSATE FOR REDUCED MARGINS**

(52) **U.S. Cl.**
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(2013.01); *G11C 16/10* (2013.01); *G11C 16/26*
(2013.01)

(71) Applicant: **SanDisk Technologies LLC**, Addison,
TX (US)

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G11C 16/0483
USPC 365/185.18, 185.29
See application file for complete search history.

(72) Inventors: **Peng Wang**, San Jose, CA (US);
Zhenni Wan, Santa Clara, CA (US);
Jia Li, San Francisco, CA (US);
Yihang Liu, Santa Clara, CA (US); **Bo
Lei**, San Ramon, CA (US)

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(73) Assignee: **Sandisk Technologies, Inc.**, Milpitas,
CA (US)

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* cited by examiner
Primary Examiner — Son T Dinh
(74) *Attorney, Agent, or Firm* — PEARL COHEN
ZEDEK LATZER BARATZ LLP

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G11C 16/04 (2006.01)
G11C 16/10 (2006.01)
G11C 16/26 (2006.01)
G11C 16/34 (2006.01)

(57) **ABSTRACT**
During a read operation for memory cells connected a
selected word line, a memory system adjusts the overdrive
voltage applied to word lines adjacent the selected word line
in order to compensate for margin degradation between the
erased data state and the lowest programmed data state.

20 Claims, 28 Drawing Sheets

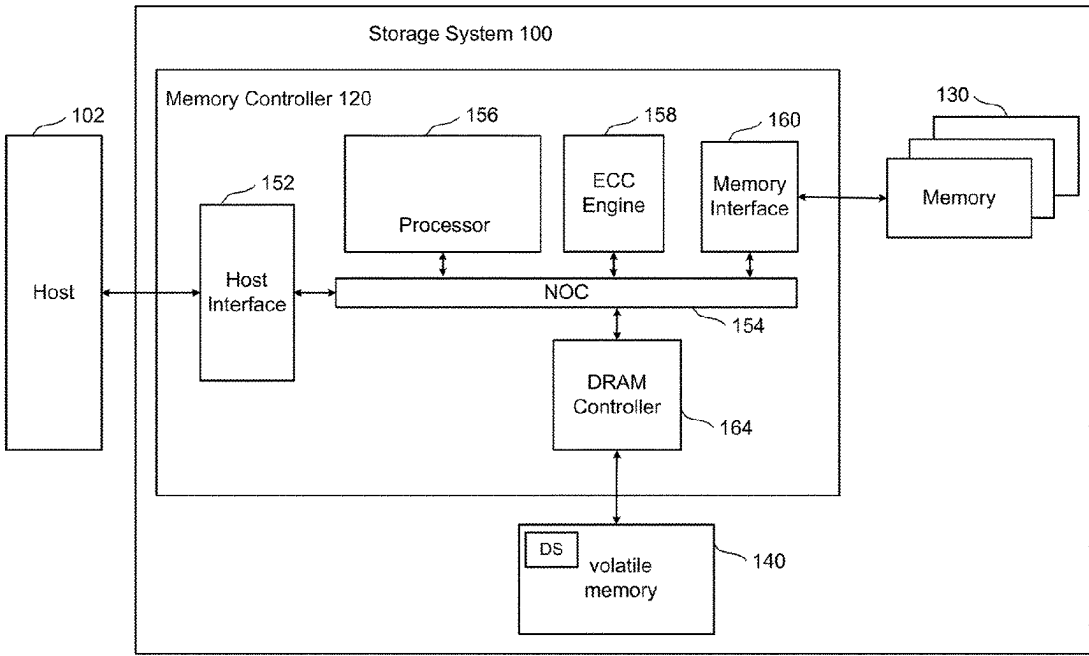
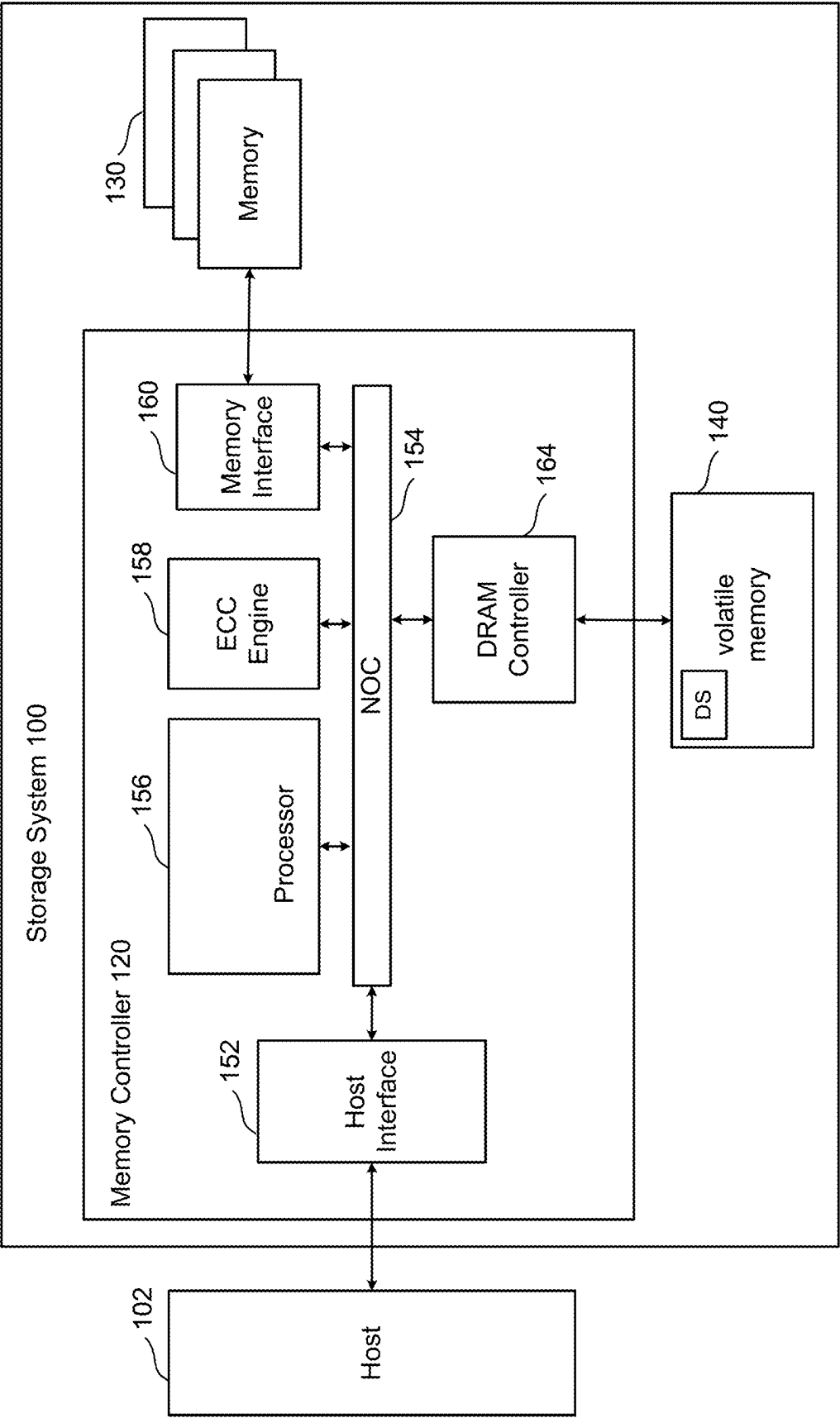
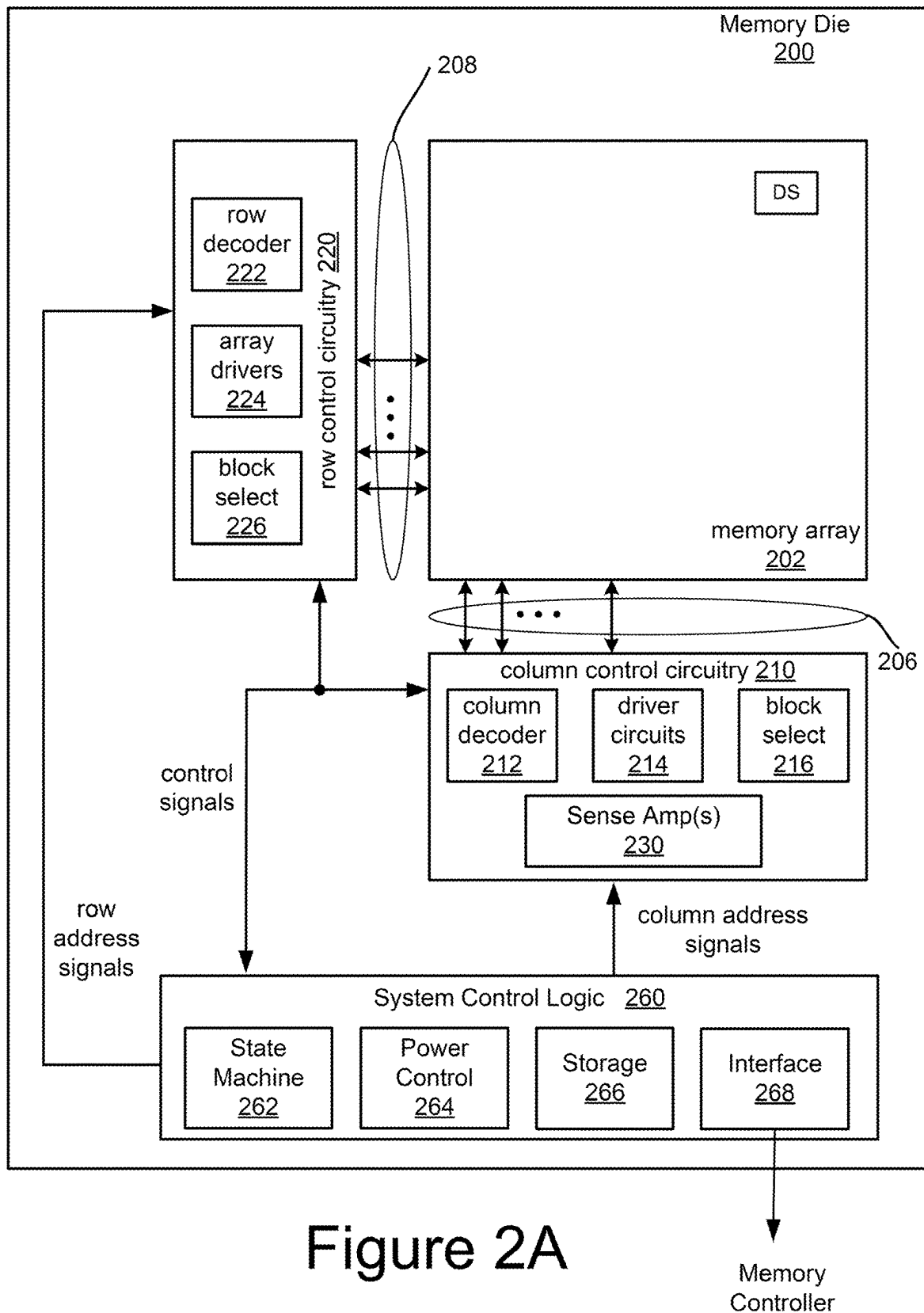


Figure 1





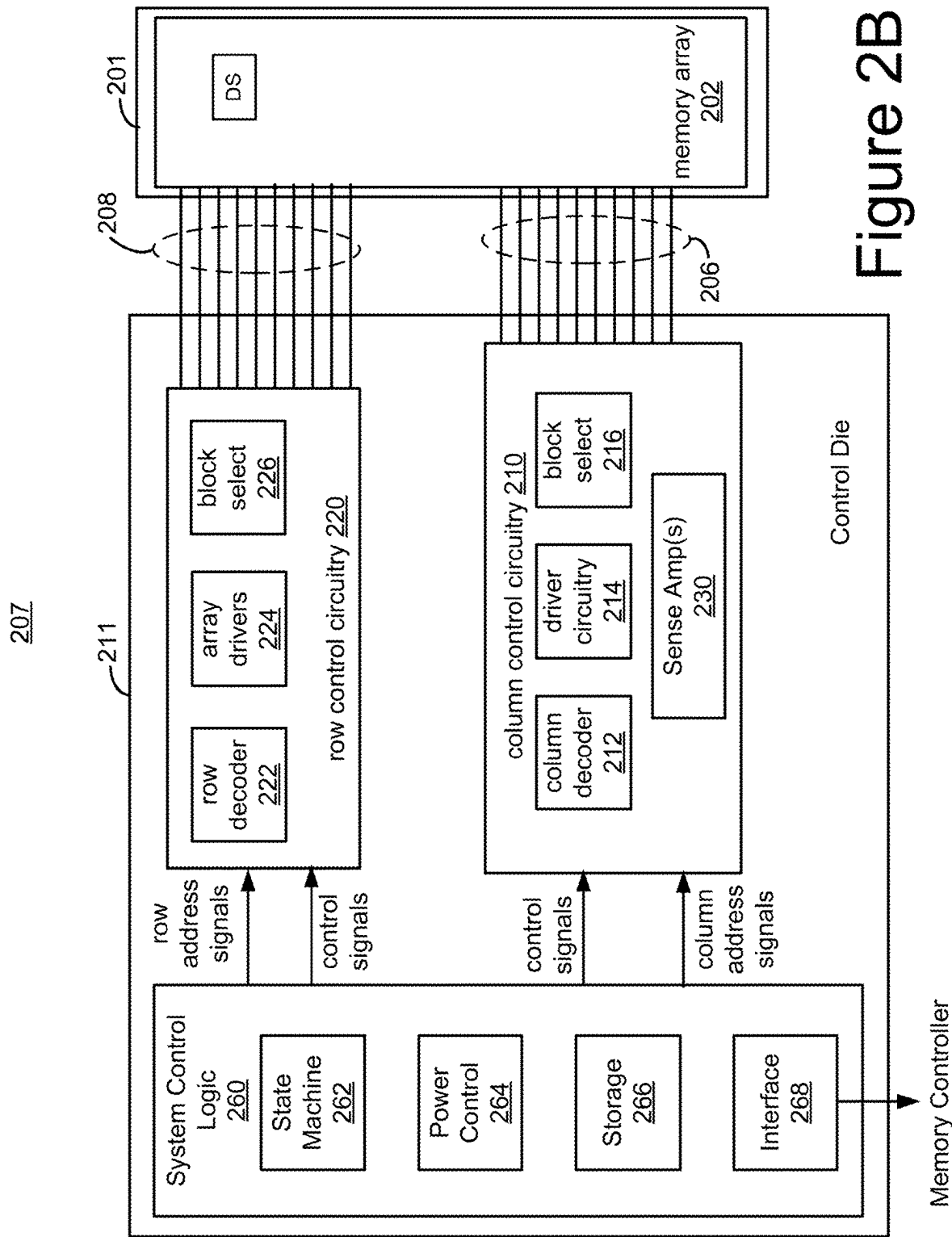


Figure 3A

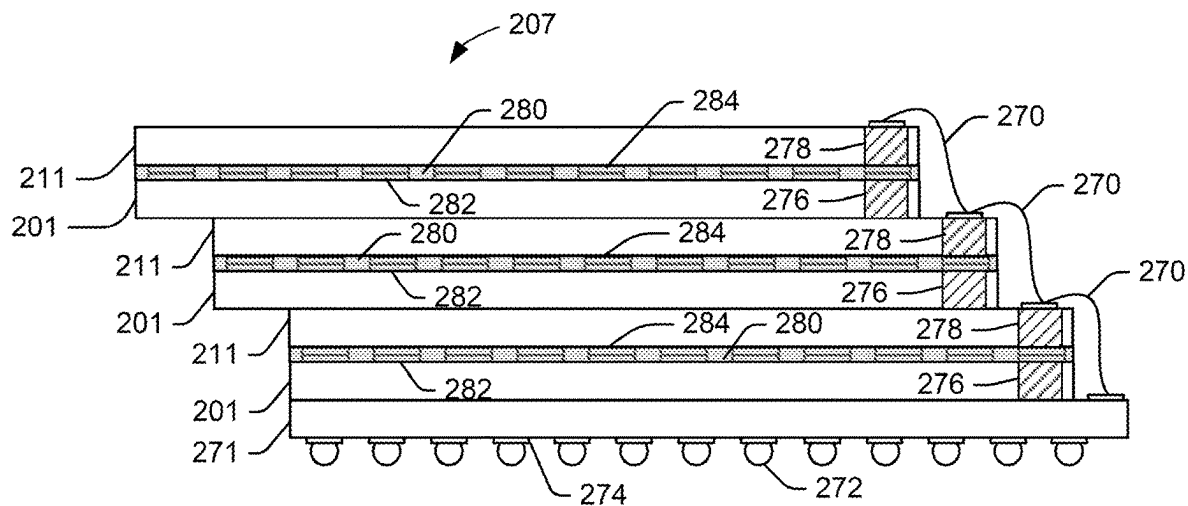


Figure 3B

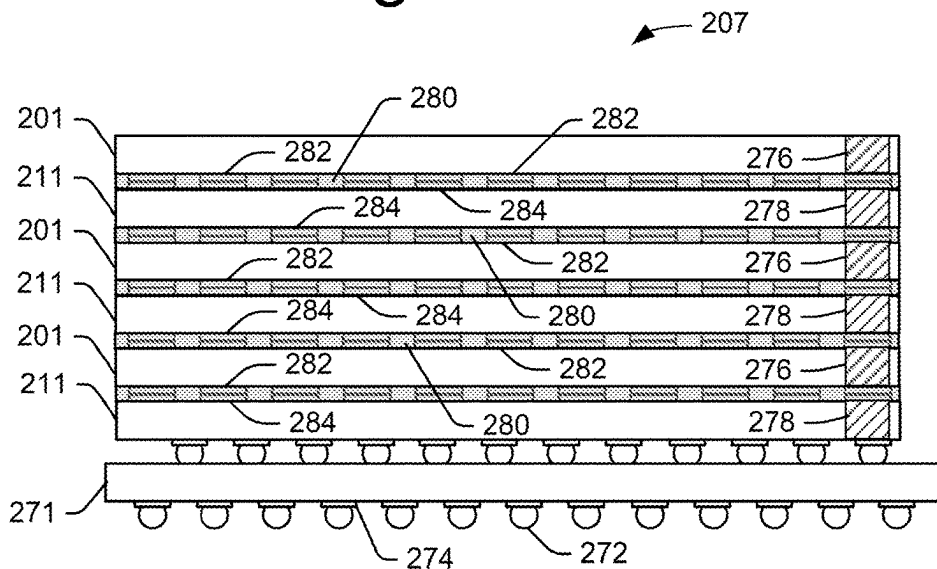


Figure 4

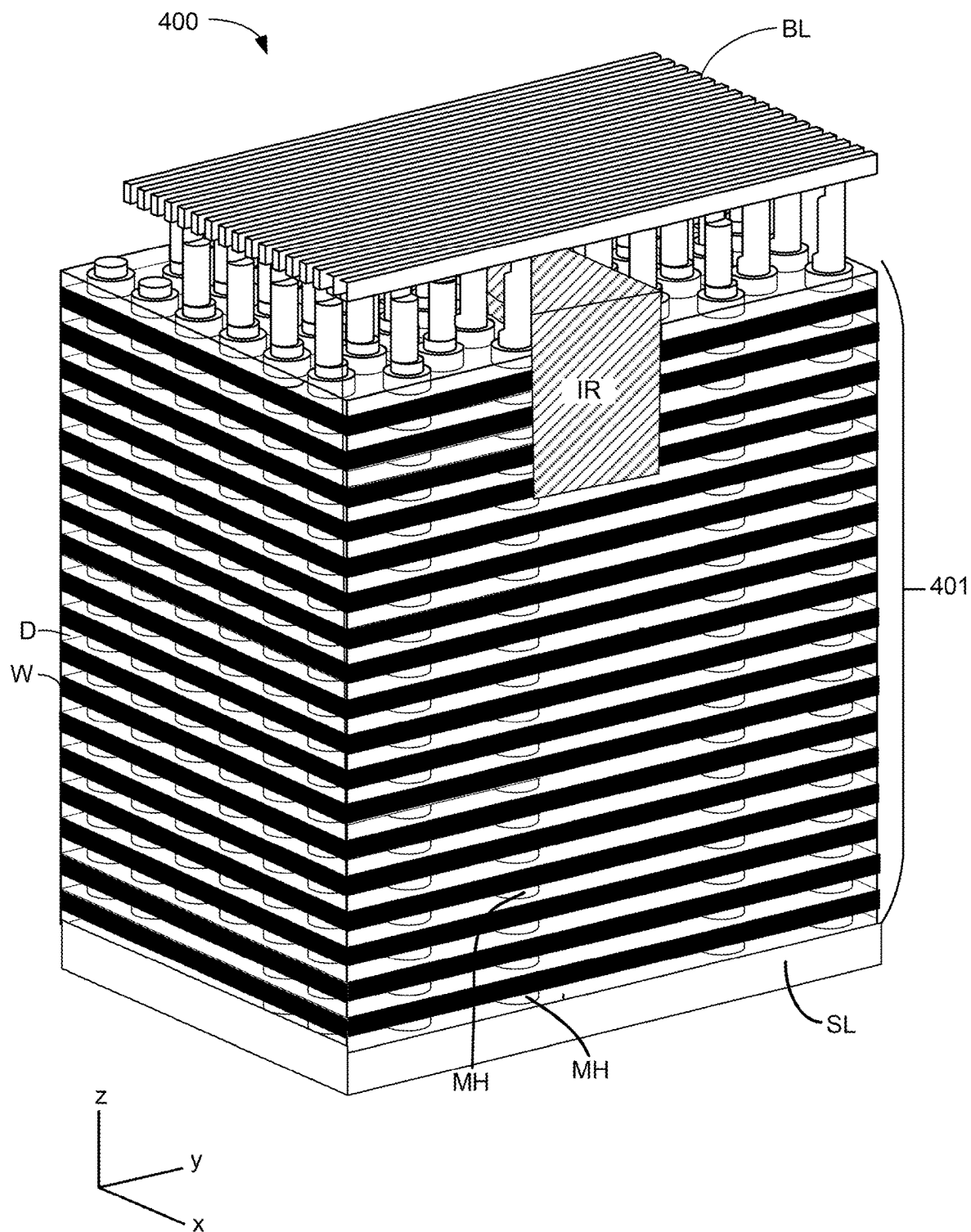


Figure 4A

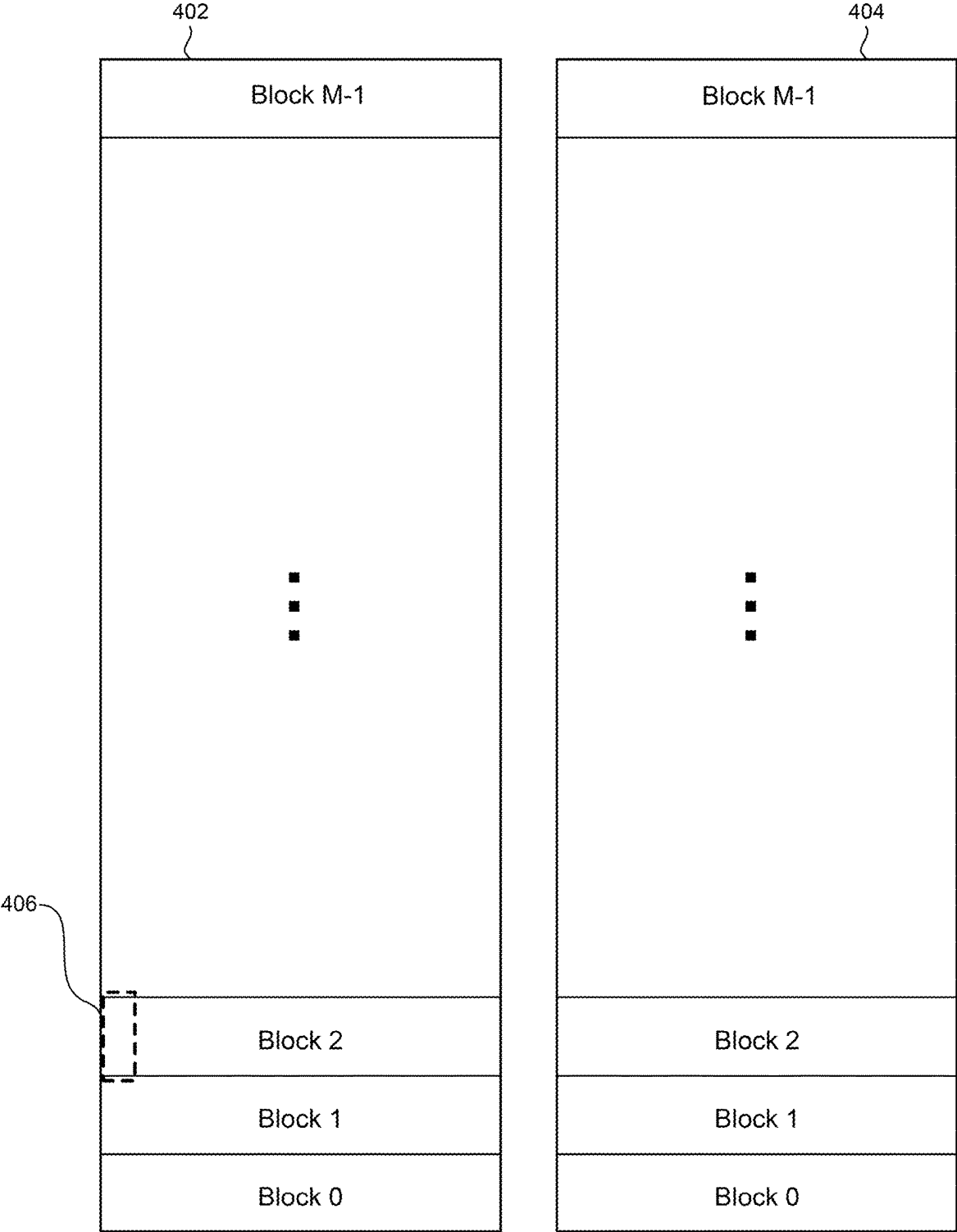


Figure 4B

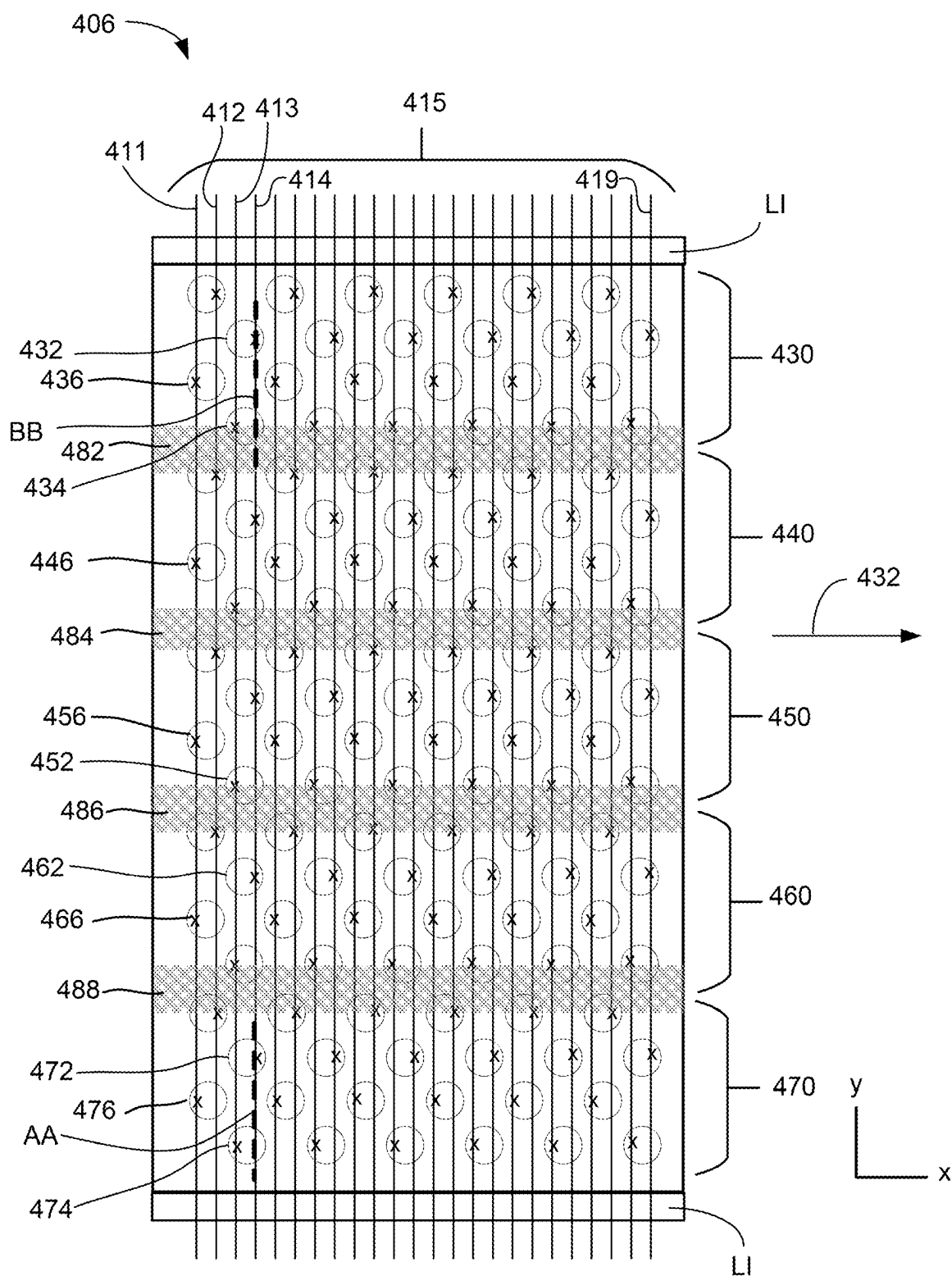


Figure 4C

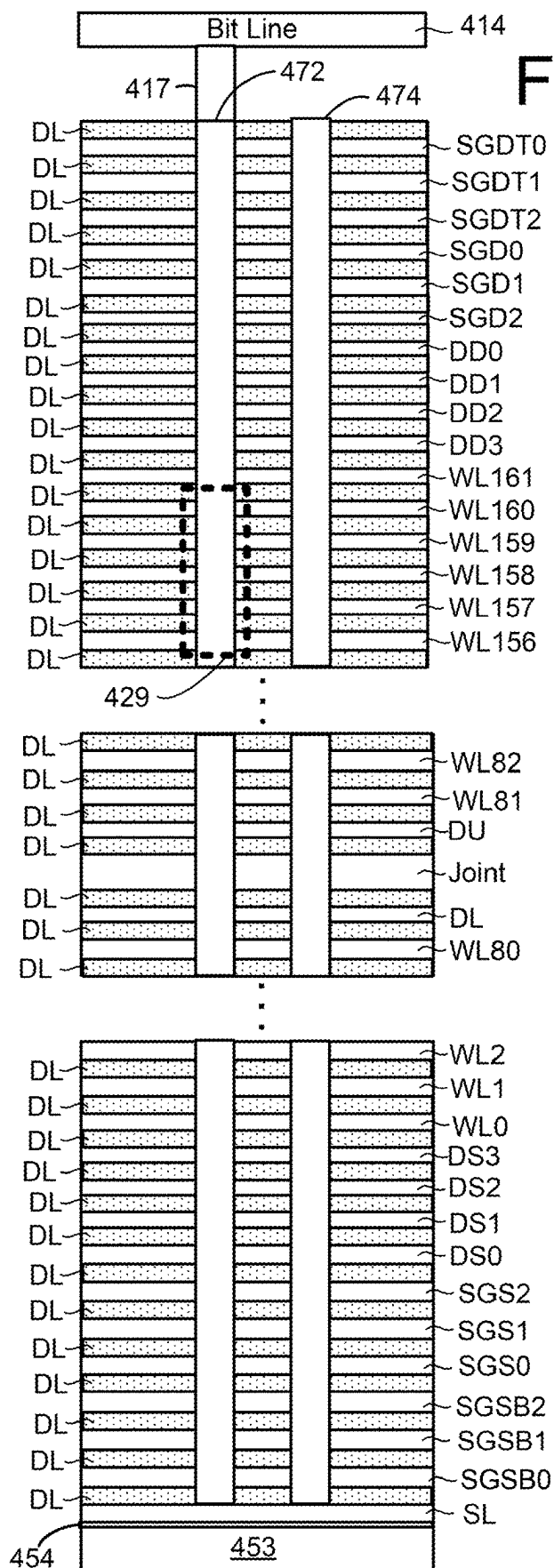
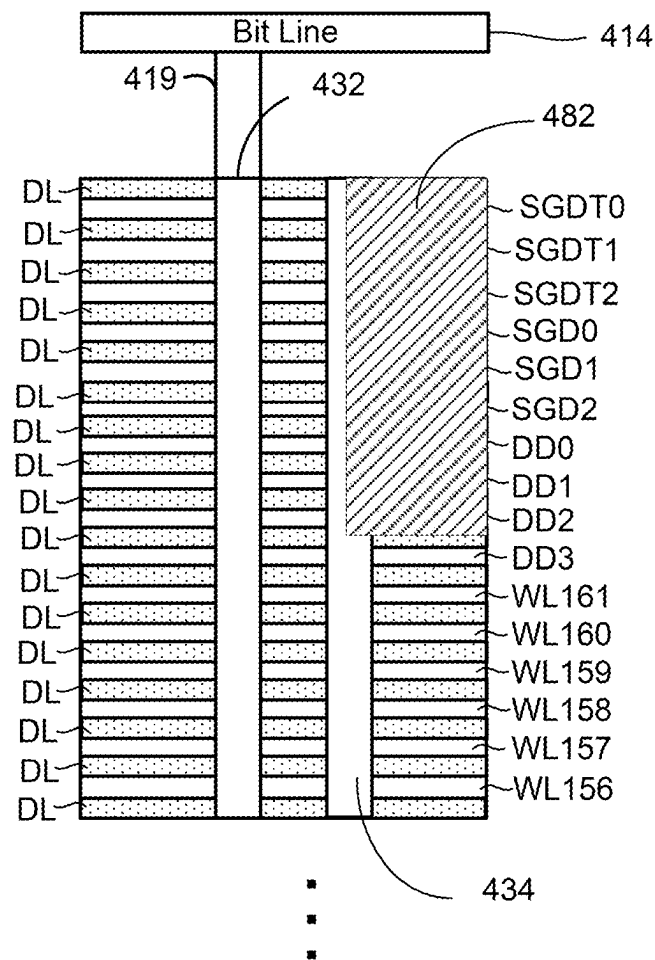


Figure 4D



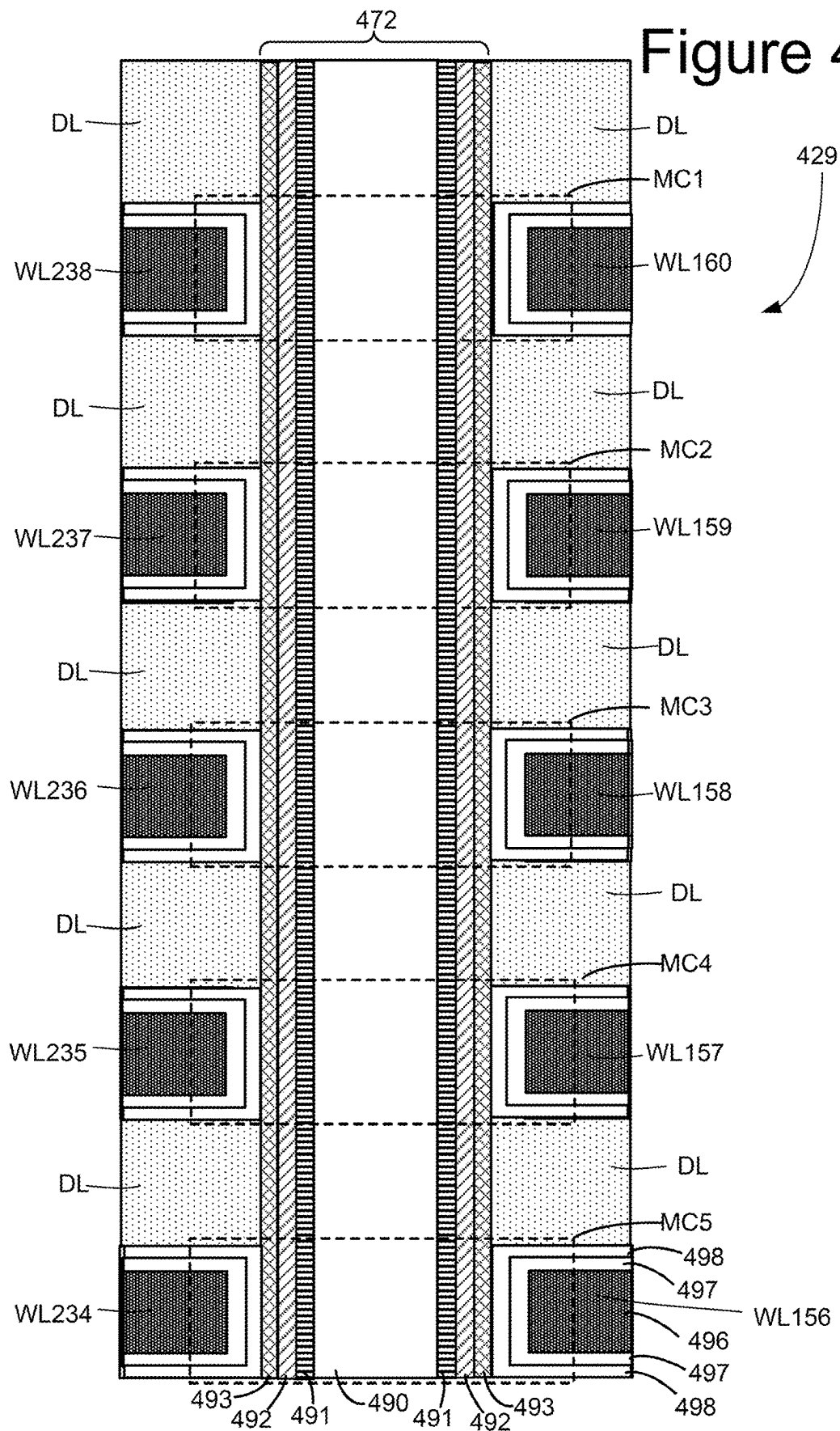


Figure 4F

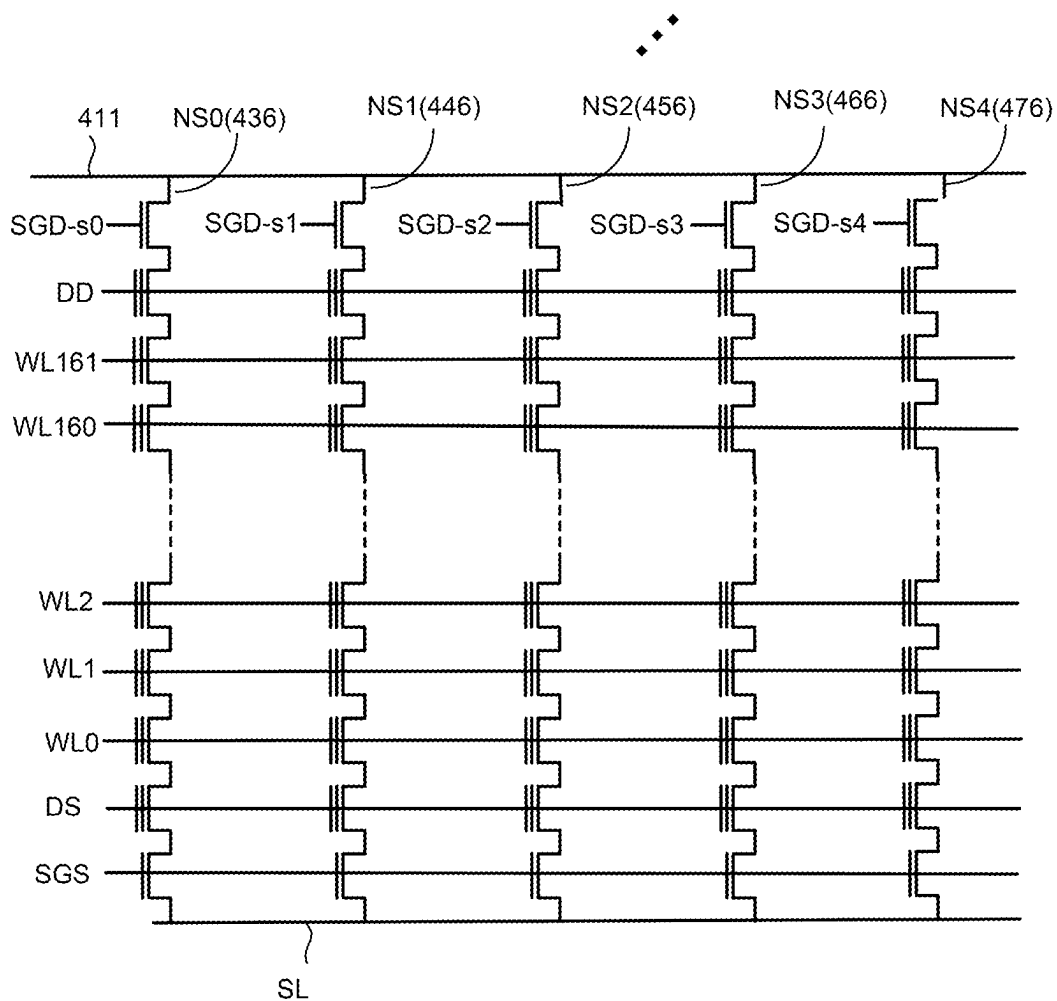


Figure 5A

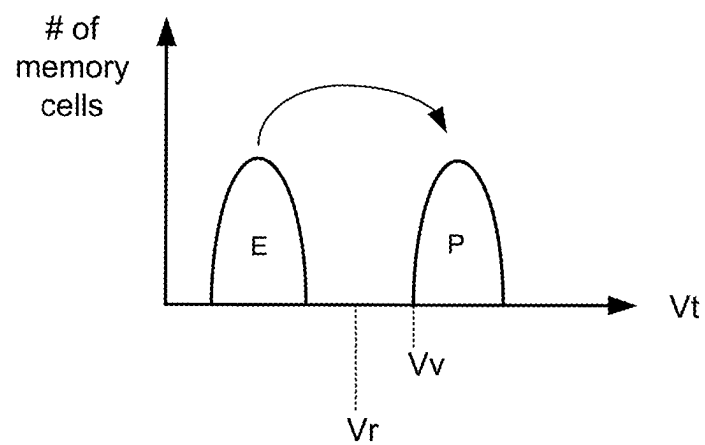


Figure 5B

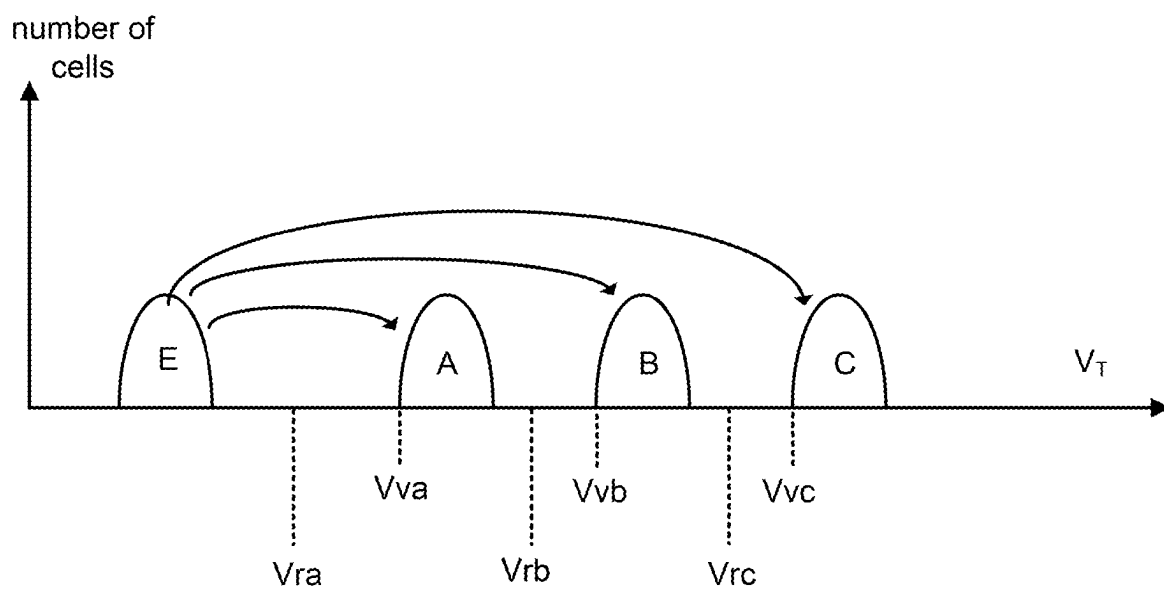


Figure 5C

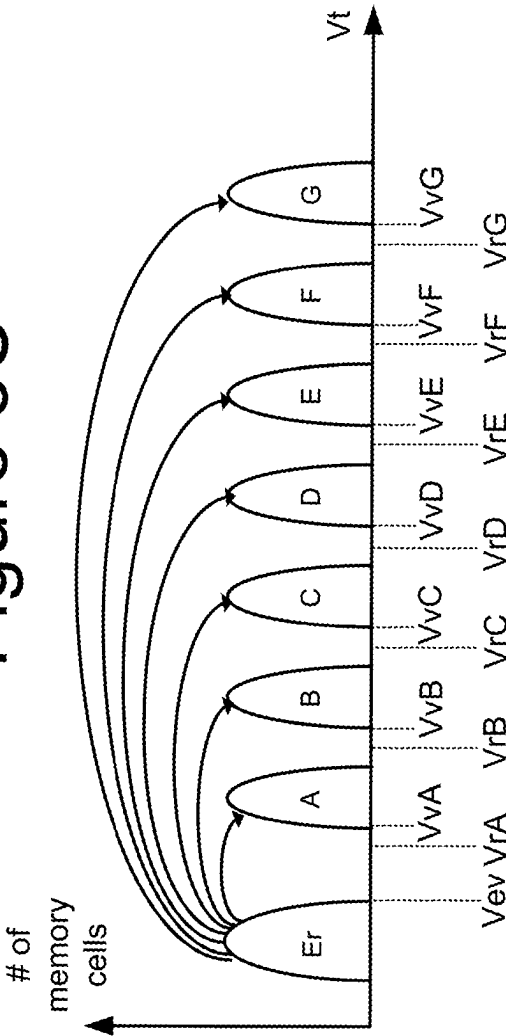


Figure 5D

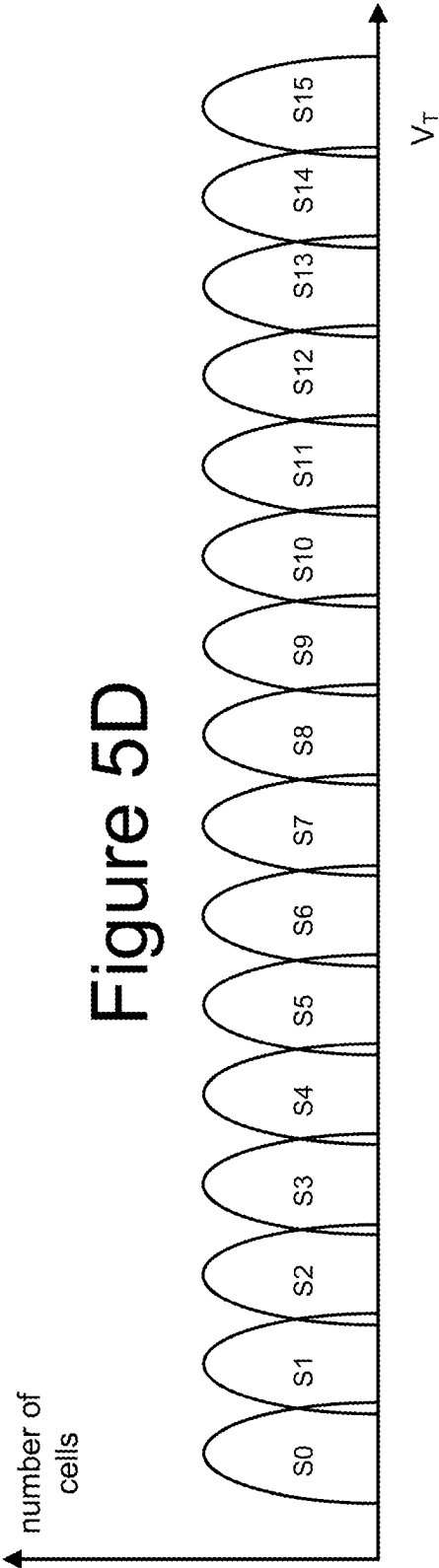


Figure 6

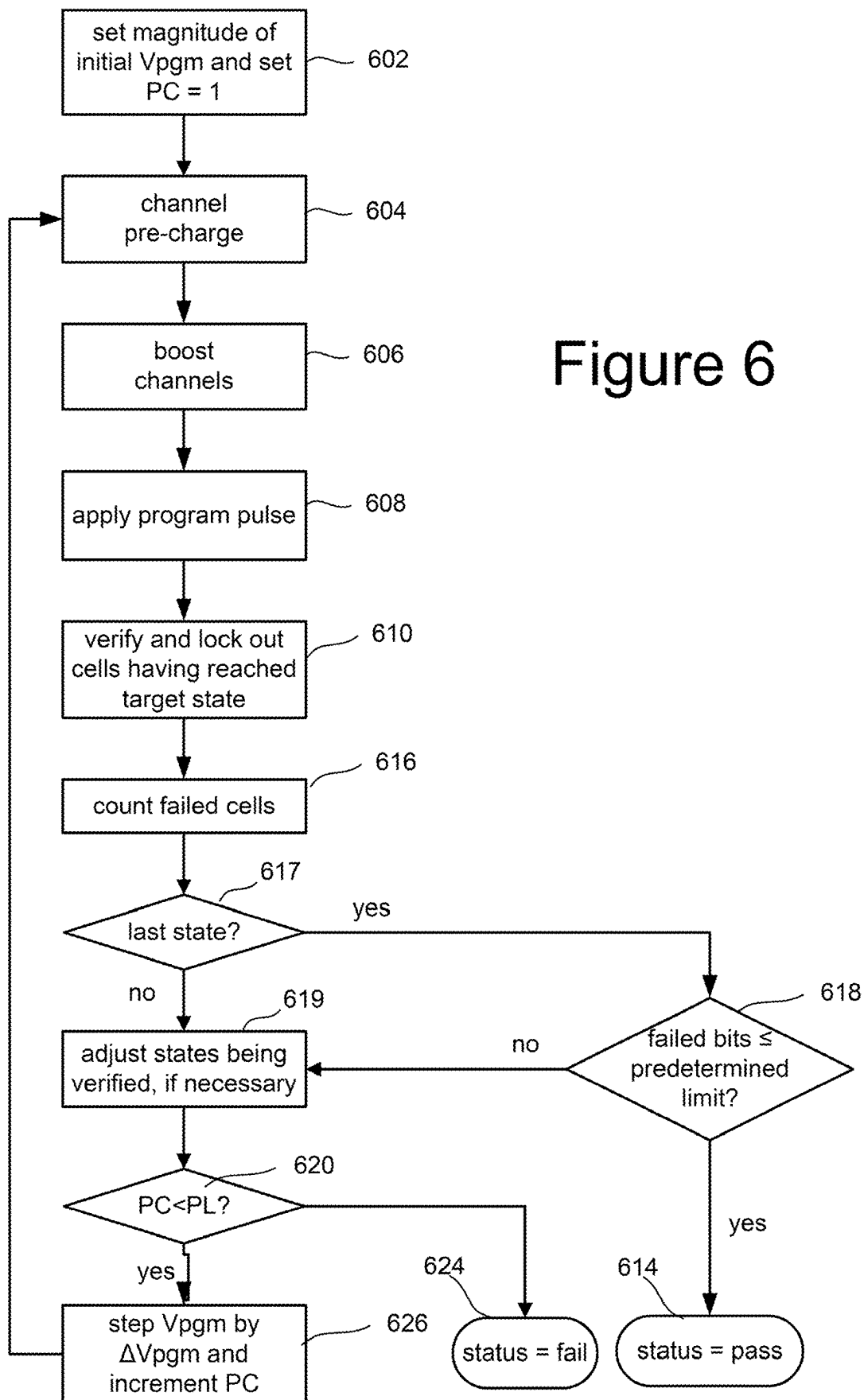


Figure 7

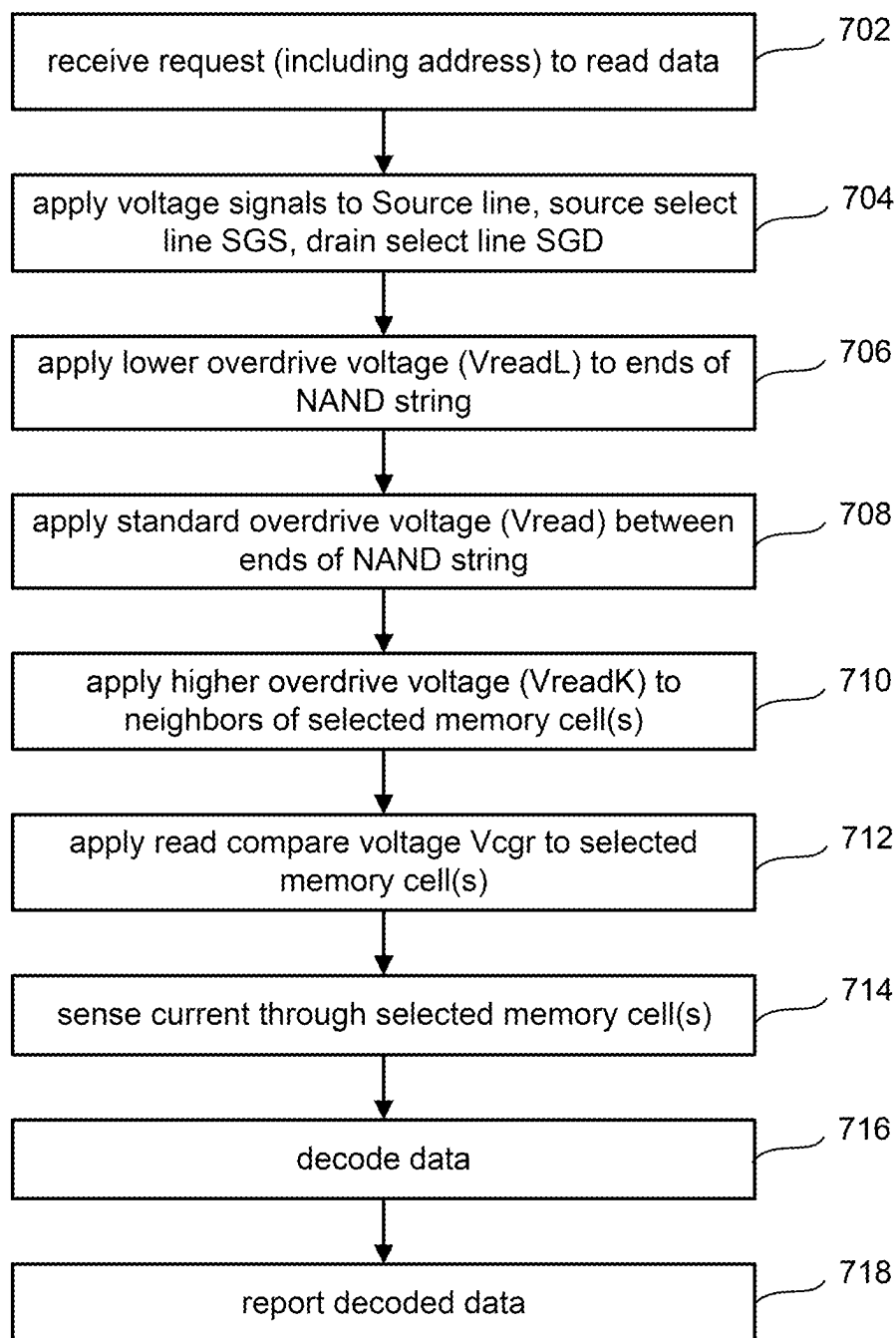


Figure 8

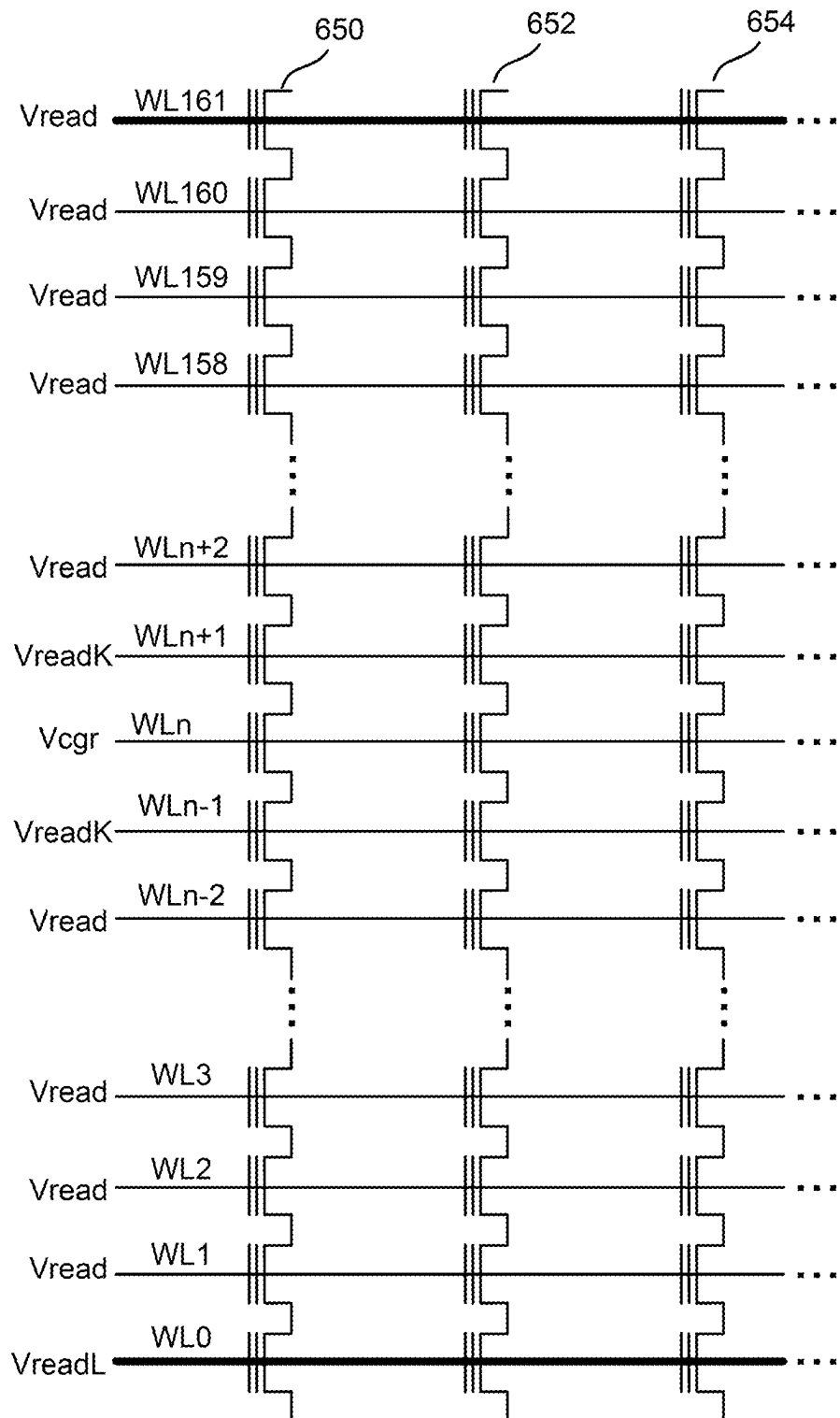


Figure 9

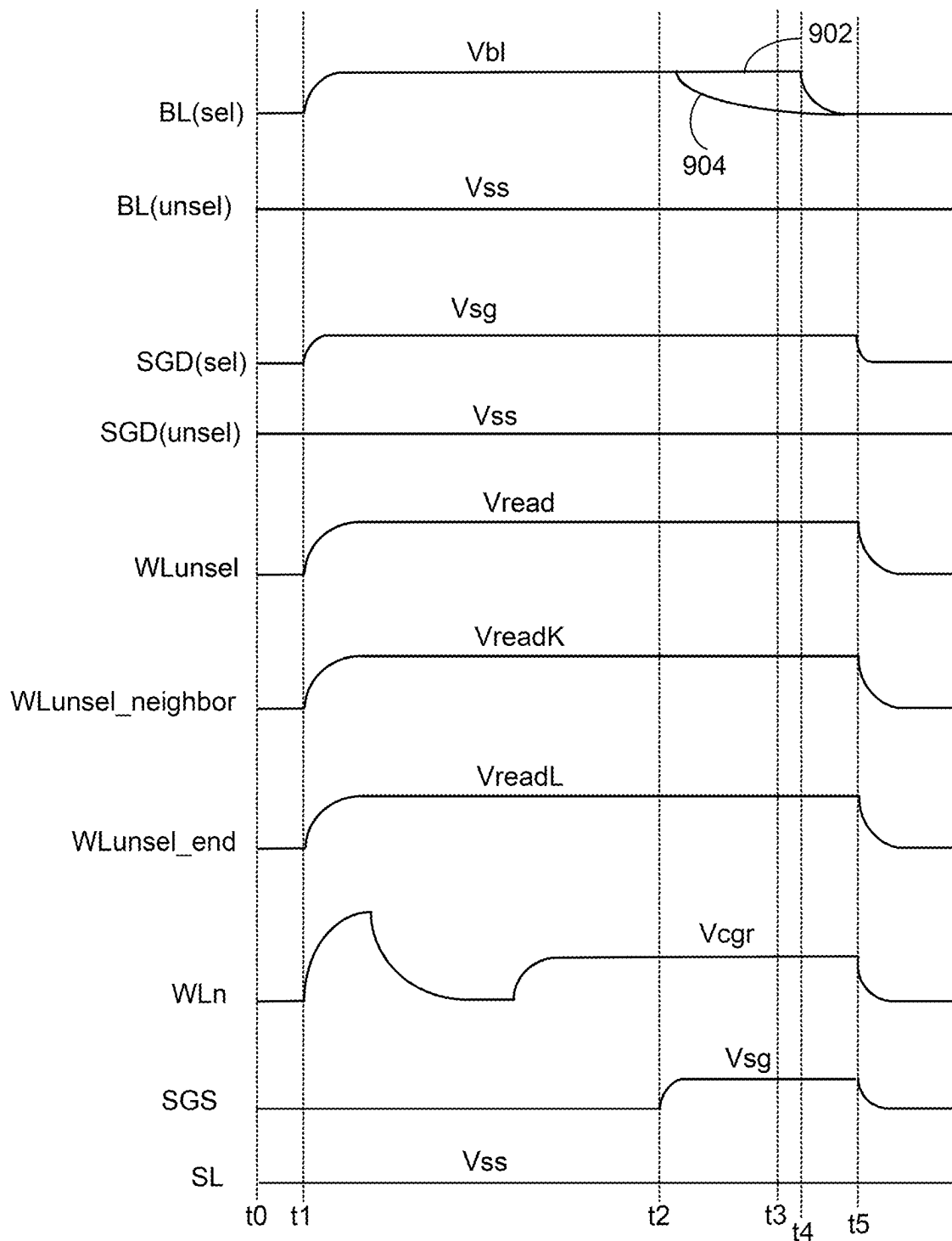


Figure 10

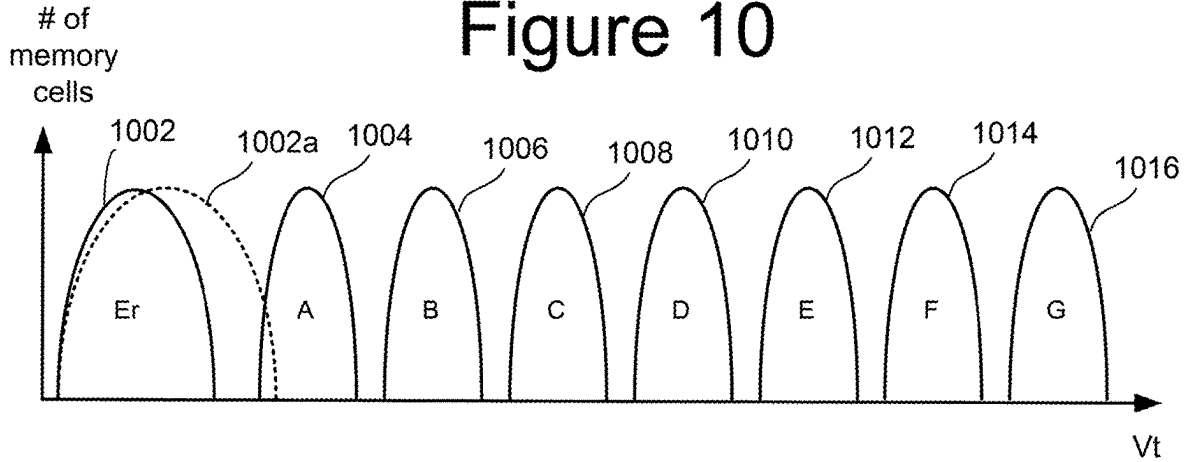


Figure 11

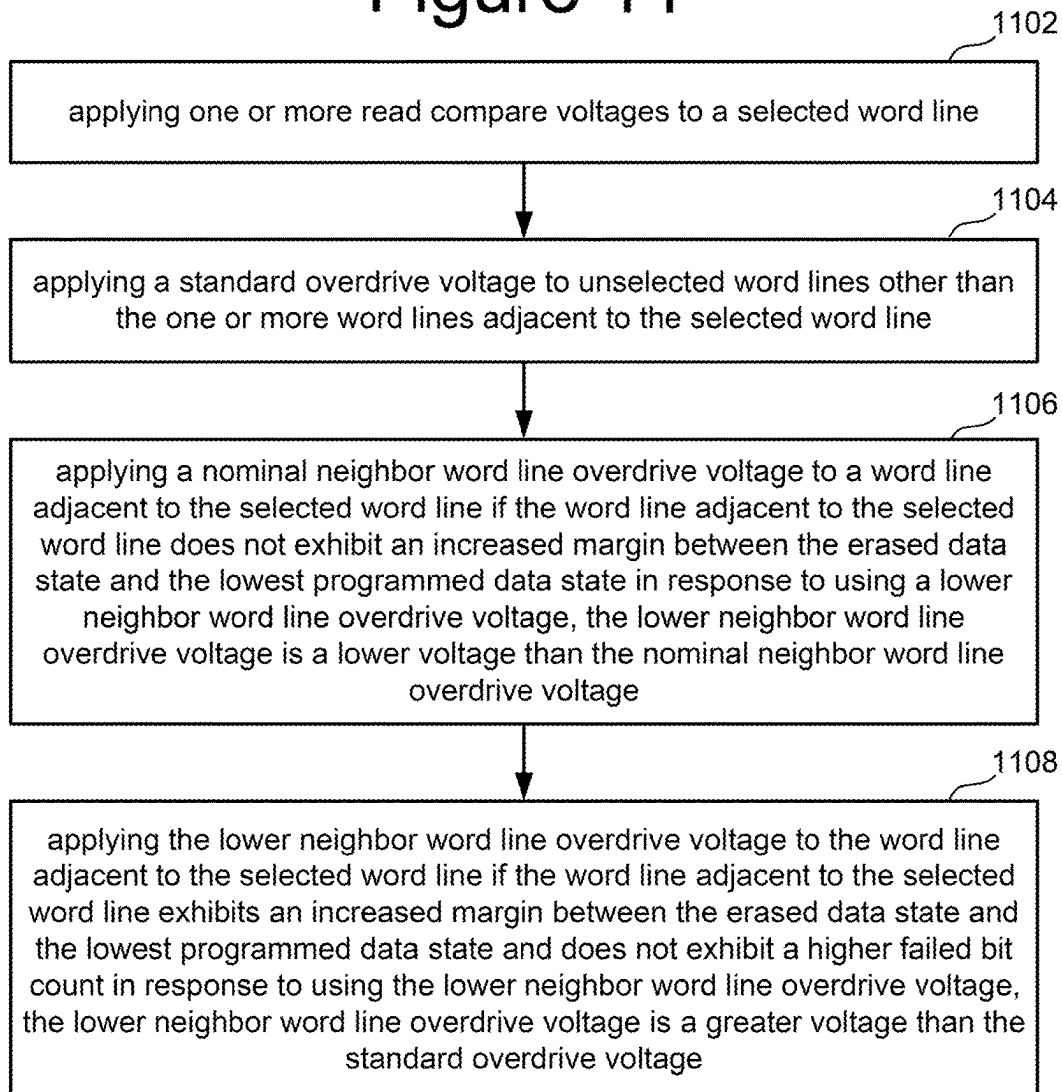


Figure 12A

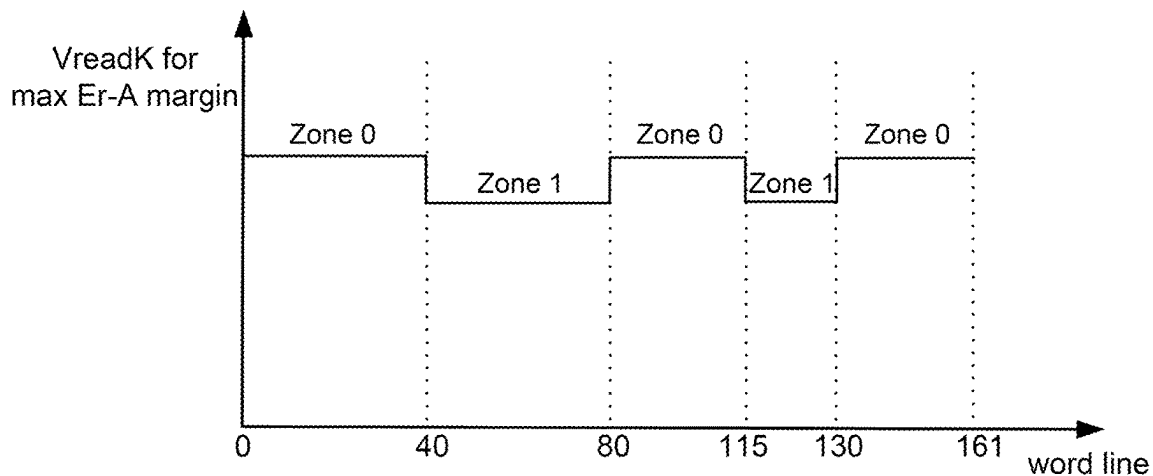


Figure 12D

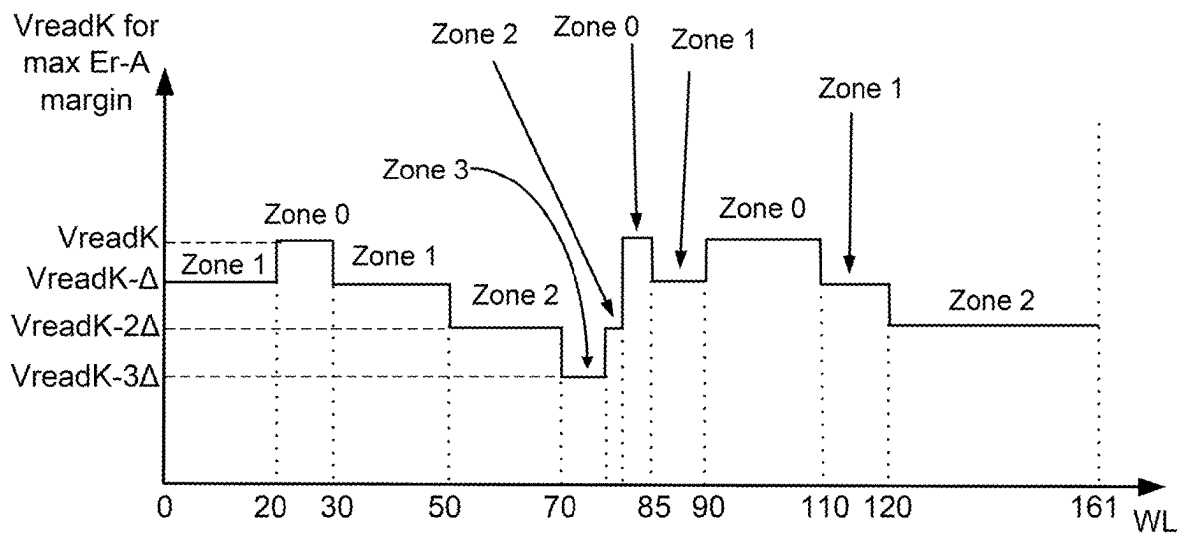


Figure 12B

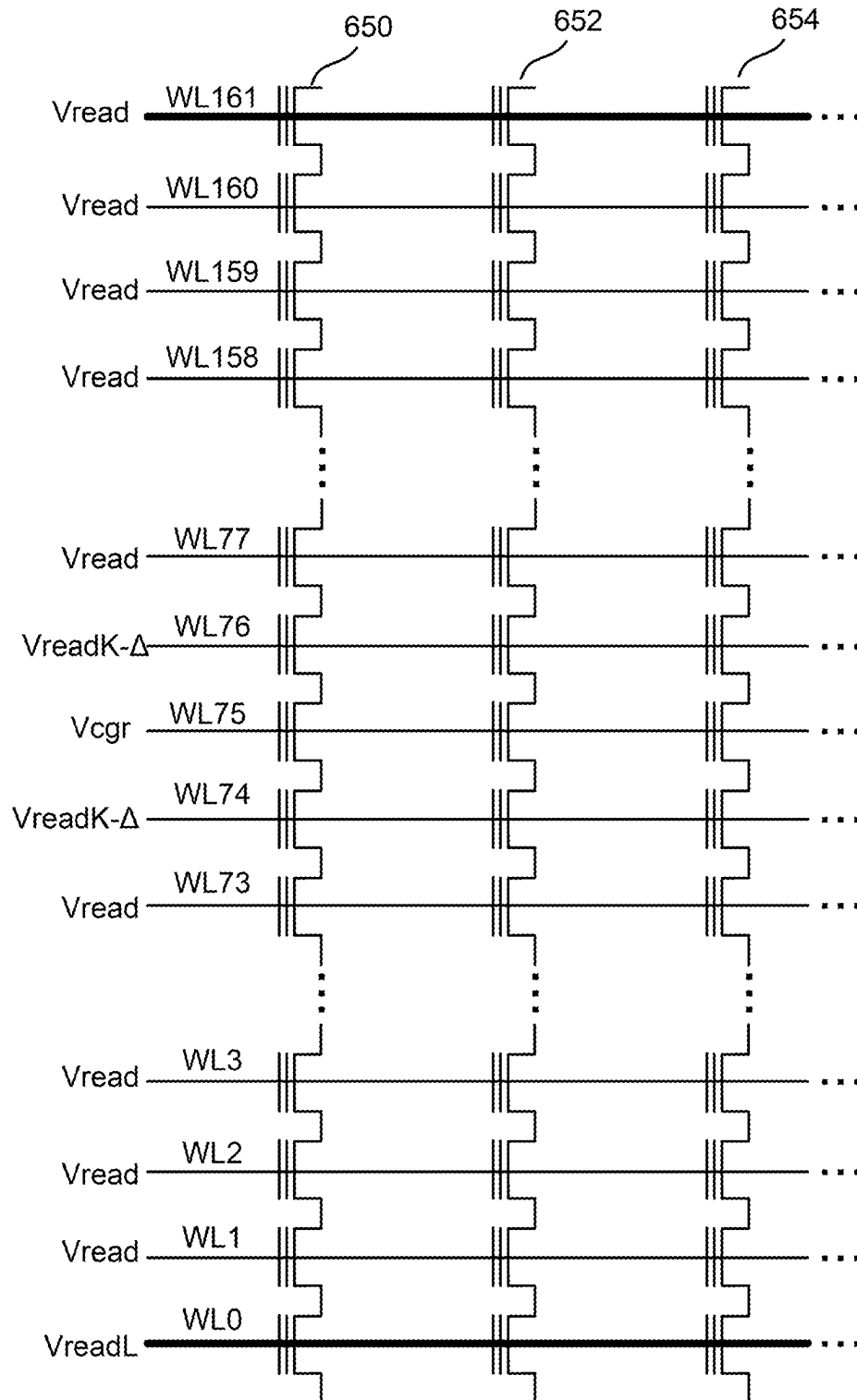


Figure 12C

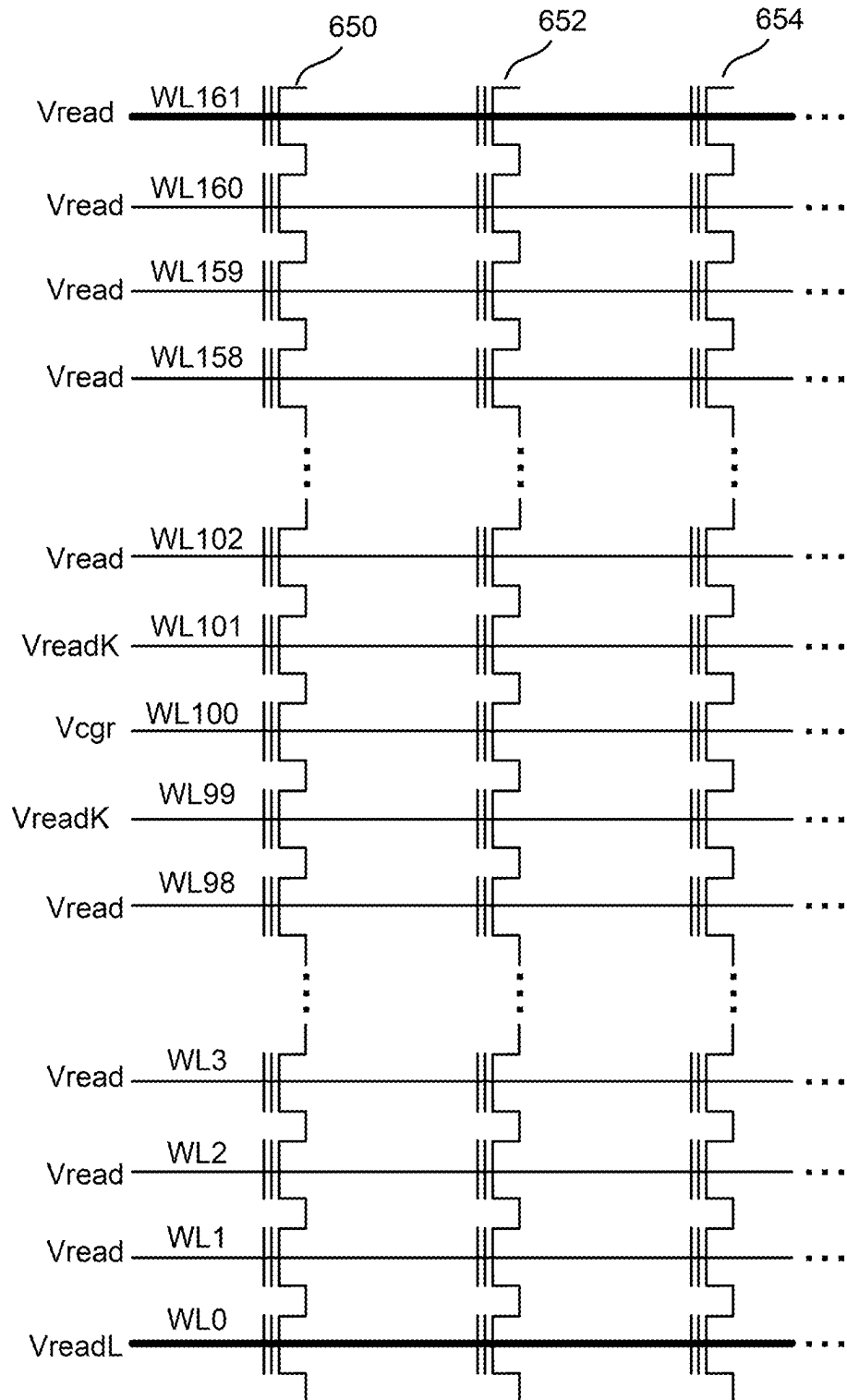


Figure 13

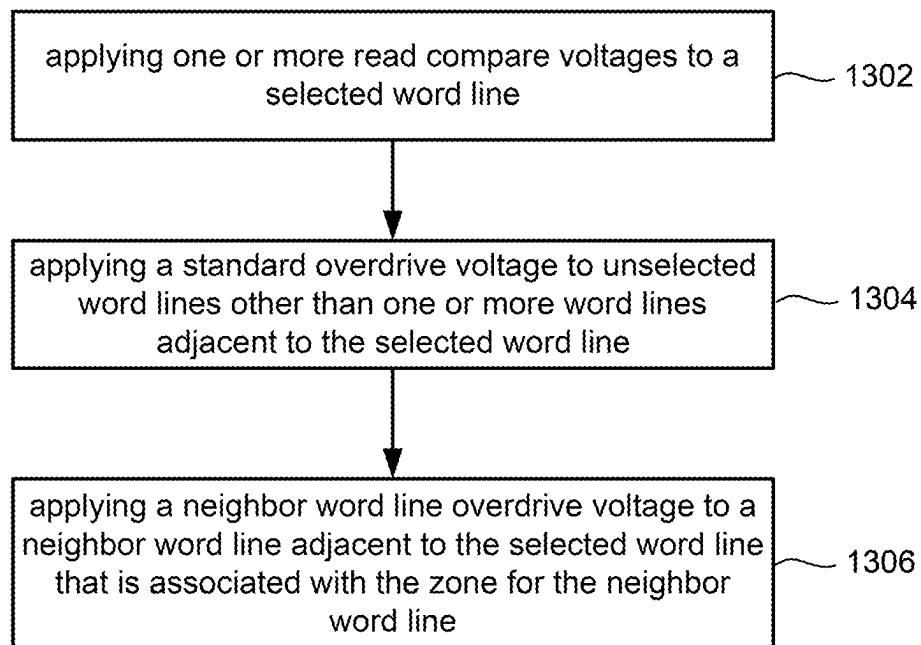


Figure 14A

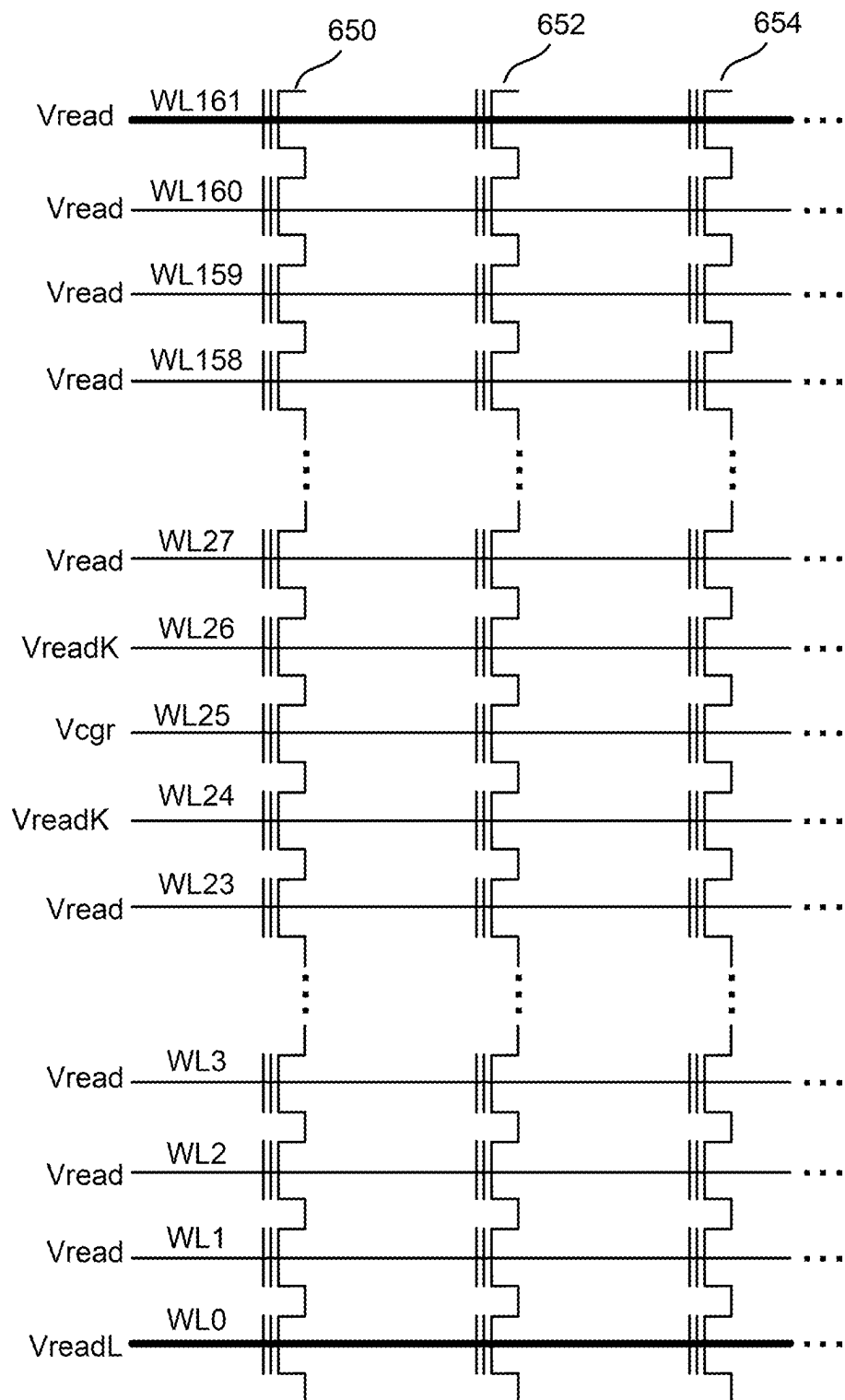


Figure 14B

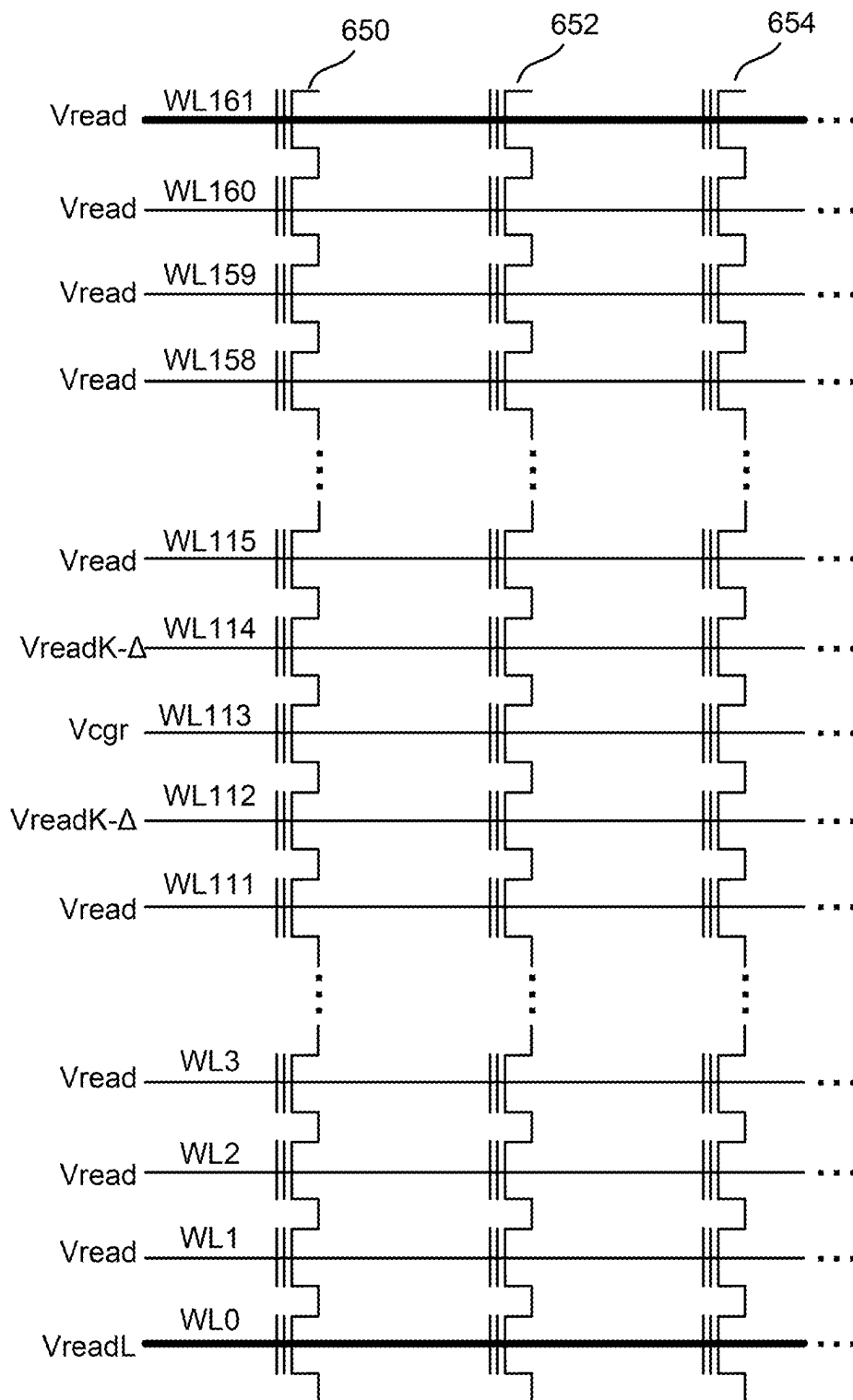


Figure 14C

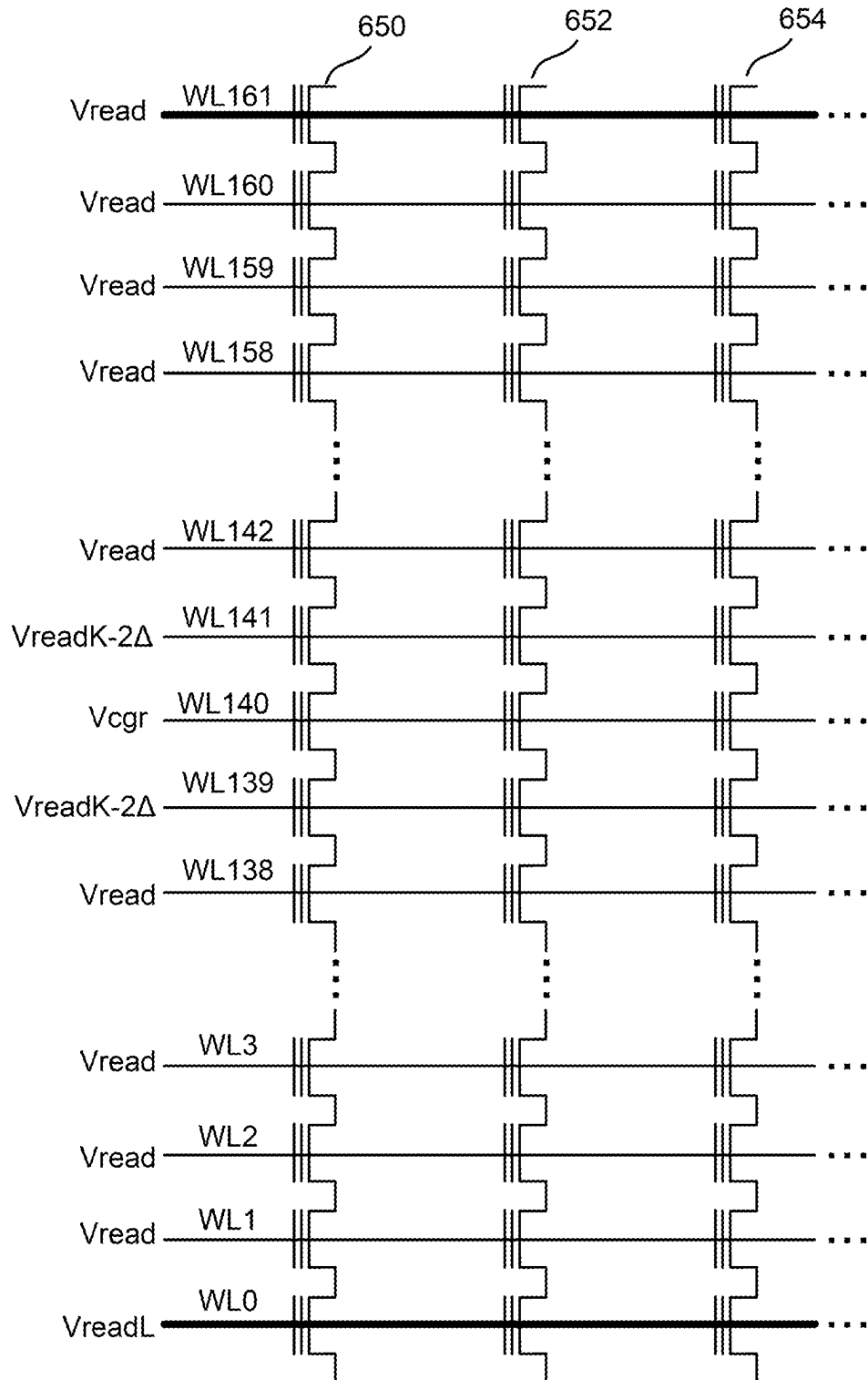


Figure 14D

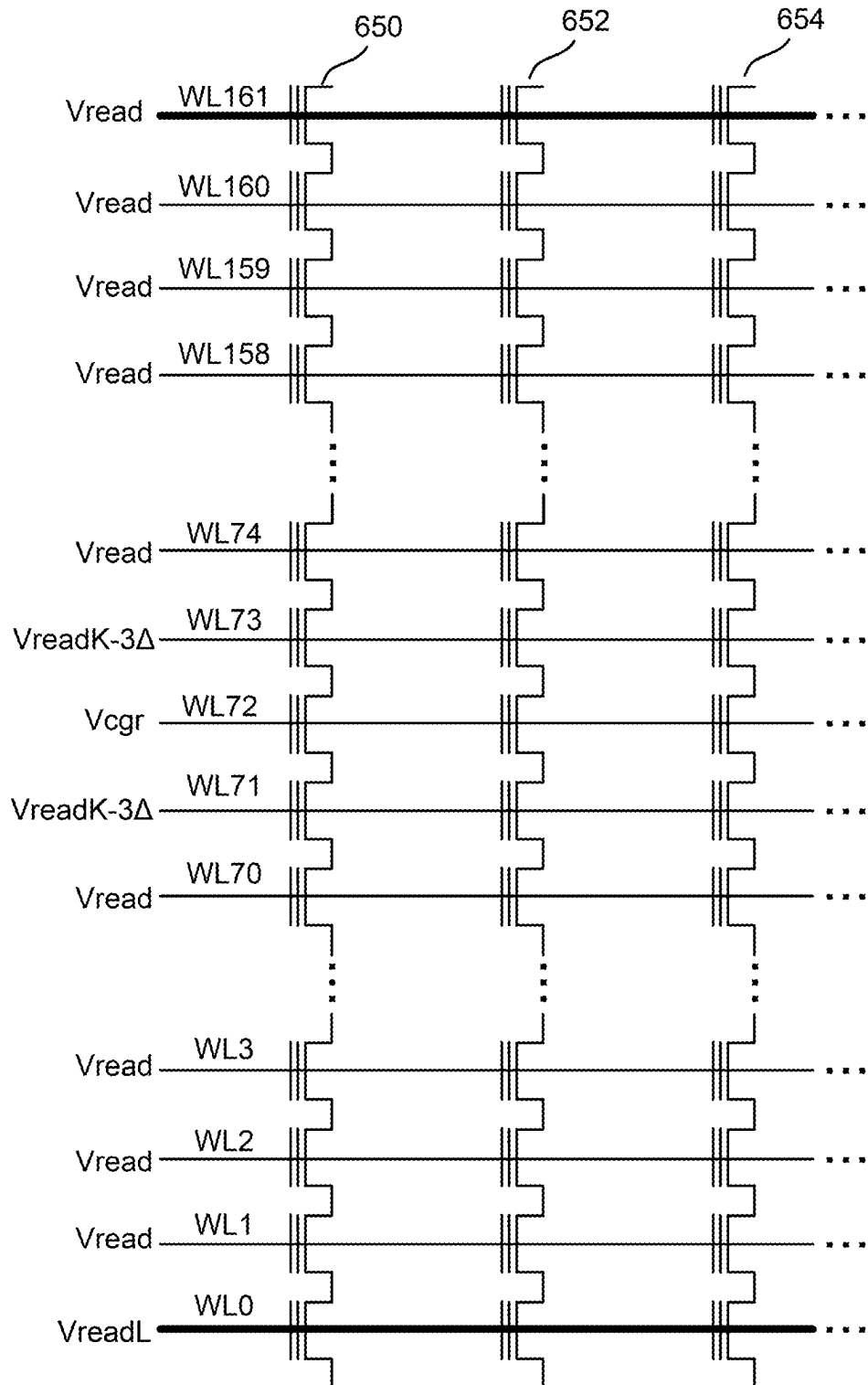


Figure 15

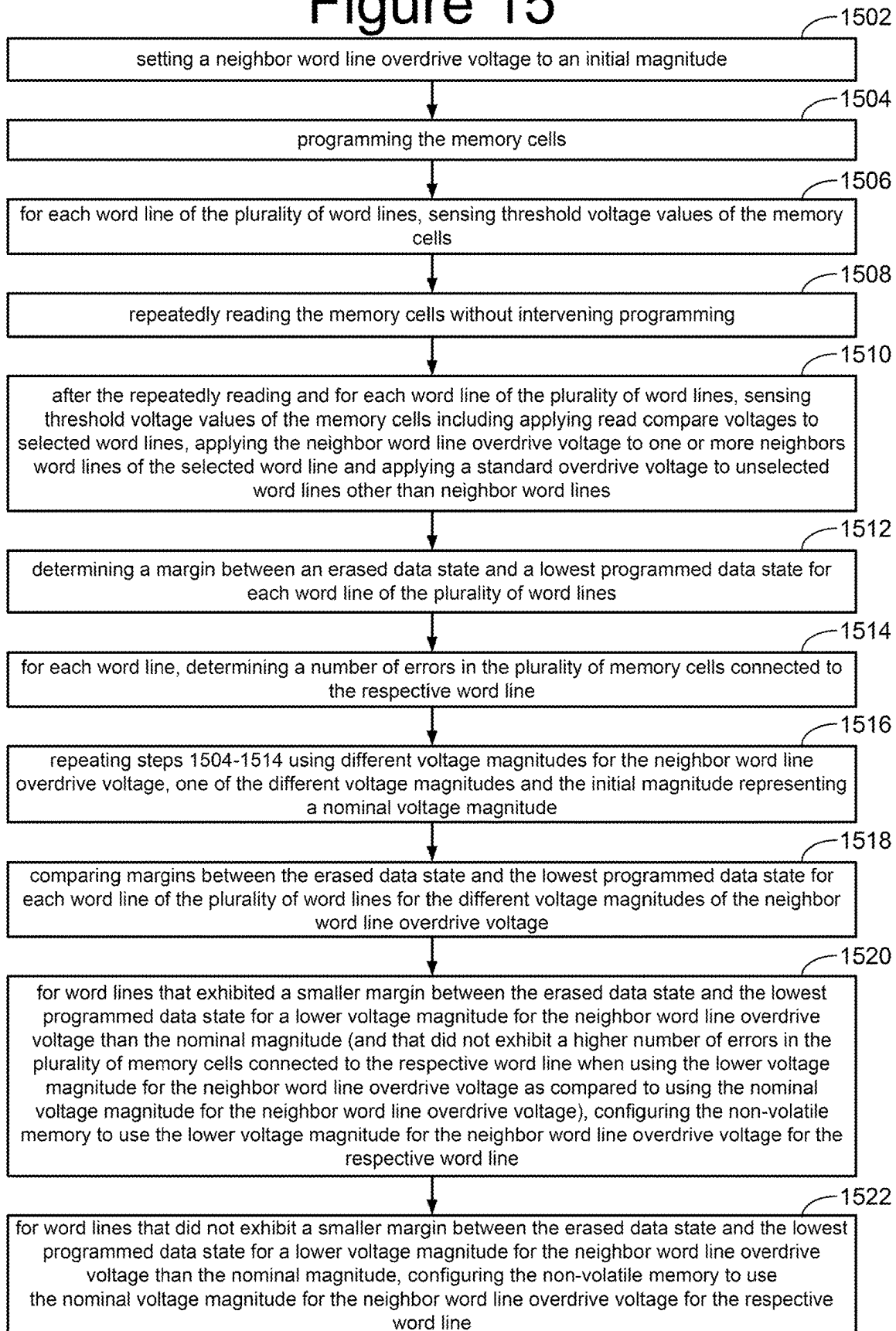


Figure 16

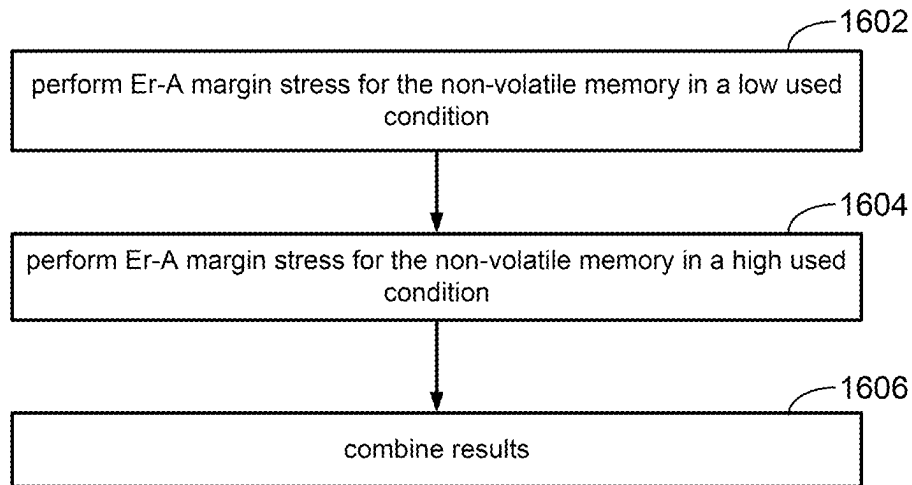
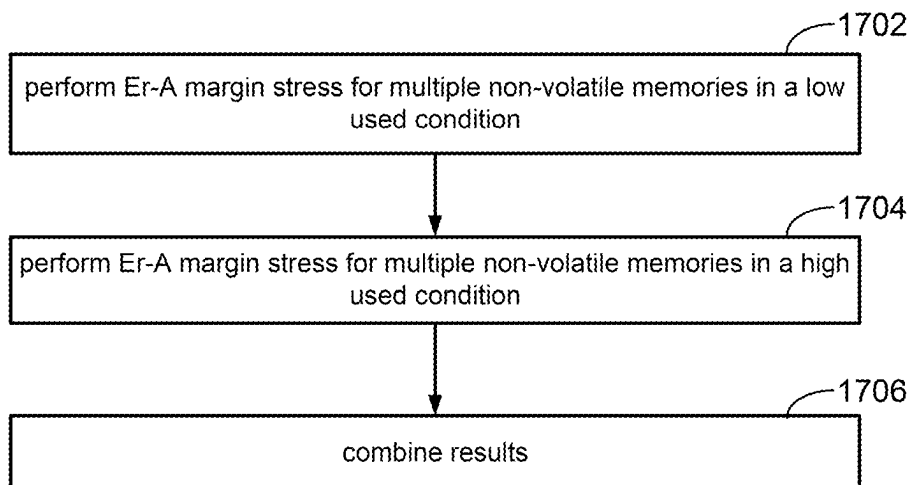


Figure 17



1

NON-VOLATILE MEMORY WITH OVERDRIVE VOLTAGE ZONING TO COMPENSATE FOR REDUCED MARGINS

CLAIM TO PRIORITY

The present application claims priority from U.S. Provisional Patent Application No. 63/415,535, entitled "NON-VOLATILE MEMORY WITH OVERDRIVE VOLTAGE ZONING TO COMPENSATE FOR REDUCED MARGINS", filed Oct. 12, 2022, incorporated by reference herein in its entirety.

BACKGROUND

The present disclosure relates to non-volatile storage.

Semiconductor memory is widely used in various electronic devices such as cellular telephones, digital cameras, personal digital assistants, medical electronics, mobile computing devices, servers, solid state drives, non-mobile computing devices and other devices. Semiconductor memory may comprise non-volatile memory or volatile memory. Non-volatile memory allows information to be stored and retained even when the non-volatile memory is not connected to a source of power (e.g., a battery). One example of non-volatile memory is flash memory (e.g., NAND-type and NOR-type flash memory).

Users of non-volatile memory can program (e.g., write) data to the non-volatile memory and later read that data back. For example, a digital camera may take a photograph and store the photograph in non-volatile memory. Later, a user of the digital camera may view the photograph by having the digital camera read the photograph from the non-volatile memory. Because users often rely on the data they store, it is important to users of non-volatile memory to be able to store data reliably so that it can be read back successfully.

BRIEF DESCRIPTION OF THE DRAWINGS

Like-numbered elements refer to common components in the different figures.

FIG. 1 is a block diagram depicting one embodiment of a storage system.

FIG. 2A is a block diagram of one embodiment of a memory die.

FIG. 2B is a block diagram of one embodiment of an integrated memory assembly.

FIGS. 3A and 3B depict different embodiments of integrated memory assemblies.

FIG. 4 is a perspective view of a portion of one embodiment of a monolithic three dimensional memory structure.

FIG. 4A is a block diagram of one embodiment of a memory structure having two planes.

FIG. 4B depicts a top view of a portion of one embodiment of a block of memory cells.

FIG. 4C depicts a cross sectional view of a portion of one embodiment of a block of memory cells.

FIG. 4D depicts a cross sectional view of a portion of one embodiment of a block of memory cells.

FIG. 4E is a cross sectional view of one embodiment of a vertical column of memory cells.

FIG. 4F is a schematic of a plurality of NAND strings in multiple regions of a same block.

FIG. 5A depicts threshold voltage distributions.

FIG. 5B depicts threshold voltage distributions.

FIG. 5C depicts threshold voltage distributions.

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FIG. 5D depicts threshold voltage distributions.

FIG. 6 is a flow chart describing one embodiment of a process for programming non-volatile memory.

FIG. 7 is a flow chart describing one embodiment of a process for reading.

FIG. 8 depicts word line voltages applied to NAND strings during a process for reading.

FIG. 9 is a timing diagram depicting voltages applied to NAND strings during a process for reading.

FIG. 10 depicts threshold voltage distributions.

FIG. 11 is a flow chart describing one embodiment of a process for reading that includes adjusting the overdrive voltage applied to word lines adjacent the selected word line in order to compensate for margin degradation between the erased data state and the lowest programmed data state.

FIG. 12A is a graph depicting the adjusting of the overdrive voltage applied to word lines adjacent the selected word line in order to compensate for margin degradation between the erased data state and the lowest programmed data state.

FIG. 12B depicts word line voltages applied to NAND strings during a process for reading.

FIG. 12C depicts word line voltages applied to NAND strings during a process for reading.

FIG. 12D is a graph depicting the adjusting of the overdrive voltage applied to word lines adjacent the selected word line in order to compensate for margin degradation between the erased data state and the lowest programmed data state.

FIG. 13 is a flow chart describing one embodiment of a process for reading that includes adjusting the overdrive voltage applied to word lines adjacent the selected word line in order to compensate for margin degradation between the erased data state and the lowest programmed data state.

FIG. 14A depicts word line voltages applied to NAND strings during a process for reading.

FIG. 14B depicts word line voltages applied to NAND strings during a process for reading.

FIG. 14C depicts word line voltages applied to NAND strings during a process for reading.

FIG. 14D depicts word line voltages applied to NAND strings during a process for reading.

FIG. 15 is a flow chart describing one embodiment of a process for configuring a non-volatile memory to adjust the overdrive voltage applied to word lines adjacent the selected word line in order to compensate for margin degradation between the erased data state and the lowest programmed data state.

FIG. 16 is a flow chart describing one embodiment of a process for configuring a non-volatile memory to adjust the overdrive voltage applied to word lines adjacent the selected word line in order to compensate for margin degradation between the erased data state and the lowest programmed data state.

FIG. 17 is a flow chart describing one embodiment of a process for configuring a non-volatile memory to adjust the overdrive voltage applied to word lines adjacent the selected word line in order to compensate for margin degradation between the erased data state and the lowest programmed data state.

DETAILED DESCRIPTION

During read operations, a read compare voltage is applied to a selected word line and overdrive voltages (standard overdrive voltage and neighbor overdrive voltage) are applied to the unselected word lines. If the overdrive volt-

ages are too high, a read disturb can occur. If the overdrive voltages are too low, the ability to discern the data state is diminished resulting in potential read errors. So there is a tradeoff for the optimum overdrive voltage. The standard overdrive voltage is applied to most unselected word lines. Neighboring word lines to the word line selected for reading receive the neighbor overdrive voltage, which is higher than the standard overdrive voltage. After many read operations, the margin between the erased state and lowest data state can become narrower. Reducing the neighbor overdrive voltage can reduce the narrowing margin issue. However, it could also lead to higher failed bit count ("FBC") during read operations. The inventors have determined that lowering the neighbor overdrive voltage for some word lines, but not all word lines, reduces the narrowing margin issue without resulting in a higher FBC. Experimentation (e.g., on actual silicon or using simulation) can be used to identify which word lines will have an increased margin between the erased state and lowest data state, and not have a higher FBC, from lowering the neighbor overdrive voltage, and then the memory can be operated with the lower neighbor overdrive voltage for those identified word lines only.

FIG. 1 is a block diagram of one embodiment of a storage system **100** that implements the proposed technology described herein. In one embodiment, storage system **100** is a solid state drive ("SSD"). Storage system **100** can also be a memory card, USB drive or other type of storage system. The proposed technology is not limited to any one type of memory system. Storage system **100** is connected to host **102**, which can be a computer, server, electronic device (e.g., smart phone, tablet or other mobile device), appliance, or another apparatus that uses memory and has data processing capabilities. In some embodiments, host **102** is separate from, but connected to, storage system **100**. In other embodiments, storage system **100** is embedded within host **102**.

The components of storage system **100** depicted in FIG. 1 are electrical circuits. Storage system **100** includes a memory controller **120** connected to non-volatile memory **130** and local high speed volatile memory **140** (e.g., DRAM). Local high speed volatile memory **140** is used by memory controller **120** to perform certain functions. For example, local high speed volatile memory **140** stores logical to physical address translation tables ("L2P tables").

Memory controller **120** comprises a host interface **152** that is connected to and in communication with host **102**. In one embodiment, host interface **152** implements a NVMe Express (NVMe) over PCI Express (PCIe). Other interfaces can also be used, such as SCSI, SATA, etc. Host interface **152** is also connected to a network-on-chip (NOC) **154**. A NOC is a communication subsystem on an integrated circuit. NOC's can span synchronous and asynchronous clock domains or use unclocked asynchronous logic. NOC technology applies networking theory and methods to on-chip communications and brings notable improvements over conventional bus and crossbar interconnections. NOC improves the scalability of systems on a chip (SoC) and the power efficiency of complex SoCs compared to other designs. The wires and the links of the NOC are shared by many signals. A high level of parallelism is achieved because all links in the NOC can operate simultaneously on different data packets. Therefore, as the complexity of integrated subsystems keep growing, a NOC provides enhanced performance (such as throughput) and scalability in comparison with previous communication architectures (e.g., dedicated point-to-point signal wires, shared buses, or segmented buses with bridges). In other embodiments, NOC **154** can be replaced by a bus. Connected to and in communication with NOC **154**

is processor **156**, ECC engine **158**, memory interface **160**, and DRAM controller **164**. DRAM controller **164** is used to operate and communicate with local high speed volatile memory **140** (e.g., DRAM). In other embodiments, local high speed volatile memory **140** can be SRAM or another type of volatile memory.

ECC engine **158** performs error correction services. For example, ECC engine **158** performs data encoding and decoding, as per the implemented ECC technique. In one embodiment, ECC engine **158** is an electrical circuit programmed by software. For example, ECC engine **158** can be a processor that can be programmed. In other embodiments, ECC engine **158** is a custom and dedicated hardware circuit without any software. In another embodiment, the function of ECC engine **158** is implemented by processor **156**.

Processor **156** performs the various controller memory operations, such as programming, erasing, reading, and memory management processes. In one embodiment, processor **156** is programmed by firmware. In other embodiments, processor **156** is a custom and dedicated hardware circuit without any software. Processor **156** also implements a translation module, as a software/firmware process or as a dedicated hardware circuit. In many systems, the non-volatile memory is addressed internally to the storage system using physical addresses associated with the one or more memory die. However, the host system will use logical addresses to address the various memory locations. This enables the host to assign data to consecutive logical addresses, while the storage system is free to store the data as it wishes among the locations of the one or more memory die. To implement this system, memory controller **120** (e.g., the translation module) performs address translation between the logical addresses used by the host and the physical addresses used by the memory dies. One example implementation is to maintain tables (i.e., the L2P tables mentioned above) that identify the current translation between logical addresses and physical addresses. An entry in the L2P table may include an identification of a logical address and corresponding physical address. Although logical address to physical address tables (or L2P tables) include the word "tables" they need not literally be tables. Rather, the logical address to physical address tables (or L2P tables) can be any type of data structure. In some examples, the memory space of a storage system is so large that the local memory **140** cannot hold all of the L2P tables. In such a case, the entire set of L2P tables are stored in a memory die **130** and a subset of the L2P tables are cached (L2P cache) in the local high speed volatile memory **140**.

Memory interface **160** communicates with non-volatile memory **130**. In one embodiment, memory interface provides a Toggle Mode interface. Other interfaces can also be used. In some example implementations, memory interface **160** (or another portion of controller **120**) implements a scheduler and buffer for transmitting data to and receiving data from one or more memory die.

In one embodiment, non-volatile memory **130** comprises one or more memory die. FIG. 2A is a functional block diagram of one embodiment of a memory die **200** that comprises non-volatile memory **130**. Each of the one or more memory die of non-volatile memory **130** can be implemented as memory die **200** of FIG. 2A. The components depicted in FIG. 2A are electrical circuits. Memory die **200** includes a memory array **202** that can comprises non-volatile memory cells, as described in more detail below. The array terminal lines of memory array **202** include the various layer(s) of word lines organized as rows, and the various layer(s) of bit lines organized as columns. However,

other orientations can also be implemented. Memory die 200 includes row control circuitry 220, whose outputs 208 are connected to respective word lines of the memory array 202. Row control circuitry 220 receives a group of M row address signals and one or more various control signals from System Control Logic circuit 260, and typically may include such circuits as row decoders 222, array terminal drivers 224, and block select circuitry 226 for both reading and writing (programming) operations. Row control circuitry 220 may also include read/write circuitry. Memory die 200 also includes column control circuitry 210 including sense amplifier(s) 230 whose input/outputs 206 are connected to respective bit lines of the memory array 202. Although only single block is shown for array 202, a memory die can include multiple arrays that can be individually accessed. Column control circuitry 210 receives a group of N column address signals and one or more various control signals from System Control Logic 260, and typically may include such circuits as column decoders 212, array terminal receivers or driver circuits 214, block select circuitry 216, as well as read/write circuitry, and I/O multiplexers.

System control logic 260 receives data and commands from memory controller 120 and provides output data and status to the host. In some embodiments, the system control logic 260 (which comprises one or more electrical circuits) include state machine 262 that provides die-level control of memory operations. In one embodiment, the state machine 262 is programmable by software. In other embodiments, the state machine 262 does not use software and is completely implemented in hardware (e.g., electrical circuits). In another embodiment, the state machine 262 is replaced by a micro-controller or microprocessor, either on or off the memory chip. System control logic 260 can also include a power control module 264 that controls the power and voltages supplied to the rows and columns of the memory structure 202 during memory operations and may include charge pumps and regulator circuit for creating regulating voltages. System control logic 260 includes storage 266 (e.g., RAM, registers, latches, etc.), which may be used to store parameters for operating the memory array 202.

Commands and data are transferred between memory controller 120 and memory die 200 via memory controller interface 268 (also referred to as a “communication interface”). Memory controller interface 268 is an electrical interface for communicating with memory controller 120. Examples of memory controller interface 268 include a Toggle Mode Interface and an Open NAND Flash Interface (ONFI). Other I/O interfaces can also be used.

In some embodiments, all the elements of memory die 200, including the system control logic 260, can be formed as part of a single die. In other embodiments, some or all of the system control logic 260 can be formed on a different die.

In one embodiment, memory structure 202 comprises a three-dimensional memory array of non-volatile memory cells in which multiple memory levels are formed above a single substrate, such as a wafer. The memory structure may comprise any type of non-volatile memory that are monolithically formed in one or more physical levels of memory cells having an active area disposed above a silicon (or other type of) substrate. In one example, the non-volatile memory cells comprise vertical NAND strings with charge-trapping layers.

In another embodiment, memory structure 202 comprises a two-dimensional memory array of non-volatile memory cells. In one example, the non-volatile memory cells are

NAND flash memory cells utilizing floating gates. Other types of memory cells (e.g., NOR-type flash memory) can also be used.

The exact type of memory array architecture or memory cell included in memory structure 202 is not limited to the examples above. Many different types of memory array architectures or memory technologies can be used to form memory structure 202. No particular non-volatile memory technology is required for purposes of the new claimed embodiments proposed herein. Other examples of suitable technologies for memory cells of the memory structure 202 include ReRAM memories (resistive random access memories), magnetoresistive memory (e.g., MRAM, Spin Transfer Torque MRAM, Spin Orbit Torque MRAM), FeRAM, phase change memory (e.g., PCM), and the like. Examples of suitable technologies for memory cell architectures of the memory structure 202 include two dimensional arrays, three dimensional arrays, cross-point arrays, stacked two dimensional arrays, vertical bit line arrays, and the like.

One example of a ReRAM cross-point memory includes reversible resistance-switching elements arranged in cross-point arrays accessed by X lines and Y lines (e.g., word lines and bit lines). In another embodiment, the memory cells may include conductive bridge memory elements. A conductive bridge memory element may also be referred to as a programmable metallization cell. A conductive bridge memory element may be used as a state change element based on the physical relocation of ions within a solid electrolyte. In some cases, a conductive bridge memory element may include two solid metal electrodes, one relatively inert (e.g., tungsten) and the other electrochemically active (e.g., silver or copper), with a thin film of the solid electrolyte between the two electrodes. As temperature increases, the mobility of the ions also increases causing the programming threshold for the conductive bridge memory cell to decrease. Thus, the conductive bridge memory element may have a wide range of programming thresholds over temperature.

Another example is magnetoresistive random access memory (MRAM) that stores data by magnetic storage elements. The elements are formed from two ferromagnetic layers, each of which can hold a magnetization, separated by a thin insulating layer. One of the two layers is a permanent magnet set to a particular polarity; the other layer's magnetization can be changed to match that of an external field to store memory. A memory device is built from a grid of such memory cells. In one embodiment for programming, each memory cell lies between a pair of write lines arranged at right angles to each other, parallel to the cell, one above and one below the cell. When current is passed through them, an induced magnetic field is created. MRAM based memory embodiments will be discussed in more detail below.

Phase change memory (PCM) exploits the unique behavior of chalcogenide glass. One embodiment uses a GeTe—Sb₂Te₃ super lattice to achieve non-thermal phase changes by simply changing the co-ordination state of the Germanium atoms with a laser pulse (or light pulse from another source). Therefore, the doses of programming are laser pulses. The memory cells can be inhibited by blocking the memory cells from receiving the light. In other PCM embodiments, the memory cells are programmed by current pulses. Note that the use of “pulse” in this document does not require a square pulse but includes a (continuous or non-continuous) vibration or burst of sound, current, voltage light, or another wave. These memory elements within the individual selectable memory cells, or bits, may include a

further series element that is a selector, such as an ovonic threshold switch or metal insulator substrate.

A person of ordinary skill in the art will recognize that the technology described herein is not limited to a single specific memory structure, memory construction or material composition, but covers many relevant memory structures within the spirit and scope of the technology as described herein and as understood by one of ordinary skill in the art.

The elements of FIG. 2A can be grouped into two parts: (1) memory structure 202 and (2) peripheral circuitry, which includes all of the other components depicted in FIG. 2A. An important characteristic of a memory circuit is its capacity, which can be increased by increasing the area of the memory die of storage system 100 that is given over to the memory structure 202; however, this reduces the area of the memory die available for the peripheral circuitry. This can place quite severe restrictions on these elements of the peripheral circuitry. For example, the need to fit sense amplifier circuits within the available area can be a significant restriction on sense amplifier design architectures. With respect to the system control logic 260, reduced availability of area can limit the available functionalities that can be implemented on-chip. Consequently, a basic trade-off in the design of a memory die for the storage system 100 is the amount of area to devote to the memory structure 202 and the amount of area to devote to the peripheral circuitry.

Another area in which the memory structure 202 and the peripheral circuitry are often at odds is in the processing involved in forming these regions, since these regions often involve differing processing technologies and the trade-off in having differing technologies on a single die. For example, when the memory structure 202 is NAND flash, this is an NMOS structure, while the peripheral circuitry is often CMOS based. For example, elements such as sense amplifier circuits, charge pumps, logic elements in a state machine, and other peripheral circuitry in system control logic 260 often employ PMOS devices. Processing operations for manufacturing a CMOS die will differ in many aspects from the processing operations optimized for an NMOS flash NAND memory or other memory cell technologies.

To improve upon these limitations, embodiments described below can separate the elements of FIG. 2A onto separately formed dies that are then bonded together. More specifically, the memory structure 202 can be formed on one die (referred to as the memory die) and some or all of the peripheral circuitry elements, including one or more control circuits, can be formed on a separate die (referred to as the control die). For example, a memory die can be formed of just the memory elements, such as the array of memory cells of flash NAND memory, MRAM memory, PCM memory, ReRAM memory, or other memory type. Some or all of the peripheral circuitry, even including elements such as decoders and sense amplifiers, can then be moved on to a separate control die. This allows each of the memory die to be optimized individually according to its technology. For example, a NAND memory die can be optimized for an NMOS based memory array structure, without worrying about the CMOS elements that have now been moved onto a control die that can be optimized for CMOS processing. This allows more space for the peripheral elements, which can now incorporate additional capabilities that could not be readily incorporated were they restricted to the margins of the same die holding the memory cell array. The two die can then be bonded together in a bonded multi-die memory circuit, with the array on the one die connected to the periphery elements on the other die. Although the following

will focus on a bonded memory circuit of one memory die and one control die, other embodiments can use more die, such as two memory die and one control die, for example.

FIG. 2B shows an alternative arrangement to that of FIG. 2A which may be implemented using wafer-to-wafer bonding to provide a bonded die pair. FIG. 2B depicts a functional block diagram of one embodiment of an integrated memory assembly 207. One or more integrated memory assemblies 207 may be used to implement the non-volatile memory 130 of storage system 100. The integrated memory assembly 207 includes two types of semiconductor die (or more succinctly, "die"). Memory die 201 includes memory structure 202. Memory structure 202 includes non-volatile memory cells. Control die 211 includes control circuitry 260, 210, and 220 (as described above). In some embodiments, control die 211 is configured to connect to the memory structure 202 in the memory die 201. In some embodiments, the memory die 201 and the control die 211 are bonded together.

FIG. 2B shows an example of the peripheral circuitry, including control circuits, formed in a peripheral circuit or control die 211 coupled to memory structure 202 formed in memory die 201. Common components are labelled similarly to FIG. 2A. System control logic 260, row control circuitry 220, and column control circuitry 210 are located in control die 211. In some embodiments, all or a portion of the column control circuitry 210 and all or a portion of the row control circuitry 220 are located on the memory die 201. In some embodiments, some of the circuitry in the system control logic 260 is located on the memory die 201.

System control logic 260, row control circuitry 220, and column control circuitry 210 may be formed by a common process (e.g., CMOS process), so that adding elements and functionalities, such as ECC, more typically found on a memory controller 120 may require few or no additional process steps (i.e., the same process steps used to fabricate controller 120 may also be used to fabricate system control logic 260, row control circuitry 220, and column control circuitry 210). Thus, while moving such circuits from a die such as memory die 201 may reduce the number of steps needed to fabricate such a die, adding such circuits to a die such as control die 211 may not require many additional process steps. The control die 211 could also be referred to as a CMOS die, due to the use of CMOS technology to implement some or all of control circuitry 260, 210, 220.

FIG. 2B shows column control circuitry 210 including sense amplifier(s) 230 on the control die 211 coupled to memory structure 202 on the memory die 201 through electrical paths 206. For example, electrical paths 206 may provide electrical connection between column decoder 212, driver circuitry 214, and block select 216 and bit lines of memory structure 202. Electrical paths may extend from column control circuitry 210 in control die 211 through pads on control die 211 that are bonded to corresponding pads of the memory die 201, which are connected to bit lines of memory structure 202. Each bit line of memory structure 202 may have a corresponding electrical path in electrical paths 206, including a pair of bond pads, which connects to column control circuitry 210. Similarly, row control circuitry 220, including row decoder 222, array drivers 224, and block select 226 are coupled to memory structure 202 through electrical paths 208. Each of electrical path 208 may correspond to a word line, dummy word line, or select gate line. Additional electrical paths may also be provided between control die 211 and memory die 201.

For purposes of this document, the phrases "a control circuit" or "one or more control circuits" can include any

one of or any combination of memory controller **120**, state machine **262**, all or a portion of system control logic **260**, all or a portion of row control circuitry **220**, all or a portion of column control circuitry **210**, a microcontroller, a microprocessor, and/or other similar functioned circuits. The control circuit can include hardware only or a combination of hardware and software (including firmware). For example, a controller programmed by firmware to perform the functions described herein is one example of a control circuit. A control circuit can include a processor, FGA, ASIC, integrated circuit, or other type of circuit.

In some embodiments, there is more than one control die **211** and more than one memory die **201** in an integrated memory assembly **207**. In some embodiments, the integrated memory assembly **207** includes a stack of multiple control die **211** and multiple memory die **201**. FIG. 3A depicts a side view of an embodiment of an integrated memory assembly **207** stacked on a substrate **271** (e.g., a stack comprising control dies **211** and memory dies **201**). The integrated memory assembly **207** has three control dies **211** and three memory dies **201**. In some embodiments, there are more than three memory dies **201** and more than three control die **211**.

Each control die **211** is affixed (e.g., bonded) to at least one of the memory dies **201**. Some of the bond pads **282/284** are depicted. There may be many more bond pads. A space between two dies **201**, **211** that are bonded together is filled with a solid layer **280**, which may be formed from epoxy or other resin or polymer. This solid layer **280** protects the electrical connections between the dies **201**, **211**, and further secures the dies together. Various materials may be used as solid layer **280**, but in embodiments, it may be Hysol epoxy resin from Henkel Corp., having offices in California, USA.

The integrated memory assembly **207** may for example be stacked with a stepped offset, leaving the bond pads at each level uncovered and accessible from above. Wire bonds **270** connected to the bond pads connect the control die **211** to the substrate **271**. A number of such wire bonds may be formed across the width of each control die **211** (i.e., into the page of FIG. 3A).

A memory die through silicon via (TSV) **276** may be used to route signals through a memory die **201**. A control die through silicon via (TSV) **278** may be used to route signals through a control die **211**. The TSVs **276**, **278** may be formed before, during or after formation of the integrated circuits in the semiconductor dies **201**, **211**. The TSVs may be formed by etching holes through the wafers. The holes may then be lined with a barrier against metal diffusion. The barrier layer may in turn be lined with a seed layer, and the seed layer may be plated with an electrical conductor such as copper, although other suitable materials such as aluminum, tin, nickel, gold, doped polysilicon, and alloys or combinations thereof may be used.

Solder balls **272** may optionally be affixed to contact pads **274** on a lower surface of substrate **271**. The solder balls **272** may be used to couple the integrated memory assembly **207** electrically and mechanically to a host device such as a printed circuit board. Solder balls **272** may be omitted where the integrated memory assembly **207** is to be used as an LGA package. The solder balls **272** may form a part of the interface between integrated memory assembly **207** and memory controller **120**.

FIG. 3B depicts a side view of another embodiment of an integrated memory assembly **207** stacked on a substrate **271**. The integrated memory assembly **207** of FIG. 3B has three control die **211** and three memory die **201**. In some embodiments, there are many more than three memory dies **201** and

many more than three control dies **211**. In this example, each control die **211** is bonded to at least one memory die **201**. Optionally, a control die **211** may be bonded to two or more memory die **201**.

Some of the bond pads **282**, **284** are depicted. There may be many more bond pads. A space between two dies **201**, **211** that are bonded together is filled with a solid layer **280**, which may be formed from epoxy or other resin or polymer. In contrast to the example in FIG. 3A, the integrated memory assembly **207** in FIG. 3B does not have a stepped offset. A memory die through silicon via (TSV) **276** may be used to route signals through a memory die **201**. A control die through silicon via (TSV) **278** may be used to route signals through a control die **211**.

Solder balls **272** may optionally be affixed to contact pads **274** on a lower surface of substrate **271**. The solder balls **272** may be used to couple the integrated memory assembly **207** electrically and mechanically to a host device such as a printed circuit board. Solder balls **272** may be omitted where the integrated memory assembly **207** is to be used as an LGA package.

As has been briefly discussed above, the control die **211** and the memory die **201** may be bonded together. Bond pads on each die **201**, **211** may be used to bond the two dies together. In some embodiments, the bond pads are bonded directly to each other, without solder or other added material, in a so-called Cu-to-Cu bonding process. In a Cu-to-Cu bonding process, the bond pads are controlled to be highly planar and formed in a highly controlled environment largely devoid of ambient particulates that might otherwise settle on a bond pad and prevent a close bond. Under such properly controlled conditions, the bond pads are aligned and pressed against each other to form a mutual bond based on surface tension. Such bonds may be formed at room temperature, though heat may also be applied. In embodiments using Cu-to-Cu bonding, the bond pads may be about 5 μm square and spaced from each other with a pitch of 5 μm to 5 μm . While this process is referred to herein as Cu-to-Cu bonding, this term may also apply even where the bond pads are formed of materials other than Cu.

When the area of bond pads is small, it may be difficult to bond the semiconductor dies together. The size of, and pitch between, bond pads may be further reduced by providing a film layer on the surfaces of the semiconductor dies including the bond pads. The film layer is provided around the bond pads. When the dies are brought together, the bond pads may bond to each other, and the film layers on the respective dies may bond to each other. Such a bonding technique may be referred to as hybrid bonding. In embodiments using hybrid bonding, the bond pads may be about 5 μm square and spaced from each other with a pitch of 1 μm to 5 μm . Bonding techniques may be used providing bond pads with even smaller (or greater) sizes and pitches.

Some embodiments may include a film on surface of the dies **201**, **211**. Where no such film is initially provided, a space between the dies may be under filled with an epoxy or other resin or polymer. The under-fill material may be applied as a liquid which then hardens into a solid layer. This under-fill step protects the electrical connections between the dies **201**, **211**, and further secures the dies together. Various materials may be used as under-fill material, but in embodiments, it may be Hysol epoxy resin from Henkel Corp., having offices in California, USA.

FIG. 4 is a perspective view of a portion of one example embodiment of a monolithic three dimensional memory array/structure that can comprise memory structure **202**, which includes a plurality non-volatile memory cells

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arranged as vertical NAND strings. For example, FIG. 4 shows a portion **400** of one block of memory. The structure depicted includes a set of bit lines BL positioned above a stack **401** of alternating dielectric layers and conductive layers. For example purposes, one of the dielectric layers is marked as D and one of the conductive layers (also called word line layers) is marked as W. The number of alternating dielectric layers and conductive layers can vary based on specific implementation requirements. As will be explained below, in one embodiment the alternating dielectric layers and conductive layers are divided into four or five (or a different number of) regions by isolation regions IR. FIG. 4 shows one isolation region IR separating two regions. Below the alternating dielectric layers and word line layers is a source line layer SL. Memory holes are formed in the stack of alternating dielectric layers and conductive layers. For example, one of the memory holes is marked as MH. Note that in FIG. 4, the dielectric layers are depicted as see-through so that the reader can see the memory holes positioned in the stack of alternating dielectric layers and conductive layers. In one embodiment, NAND strings are formed by filling the memory hole with materials including a charge-trapping material to create a vertical column of memory cells. Each memory cell can store one or more bits of data. Thus, the non-volatile memory cells are arranged in memory holes. More details of the three dimensional monolithic memory array that comprises memory structure **202** is provided below.

FIG. 4A is a block diagram explaining one example organization of memory structure **202**, which is divided into two planes **402** and **404**. Each plane is then divided into M blocks. In one example, each plane has about 2000 blocks. However, different numbers of blocks and planes can also be used. In one embodiment, a block of memory cells is a unit of erase. That is, all memory cells of a block are erased together. In other embodiments, blocks can be divided into sub-blocks and the sub-blocks can be the unit of erase. Memory cells can also be grouped into blocks for other reasons, such as to organize the memory structure to enable the signaling and selection circuits. In some embodiments, a block represents a groups of connected memory cells as the memory cells of a block share a common set of word lines. For example, the word lines for a block are all connected to all of the vertical NAND strings for that block. Although FIG. 4A shows two planes **402/404**, more or less than two planes can be implemented. In some embodiments, memory structure **202** includes eight planes.

FIGS. 4B-4G depict an example three dimensional ("3D") NAND structure that corresponds to the structure of FIG. 4 and can be used to implement memory structure **202** of FIGS. 2A and 2B. FIG. 4B is a block diagram depicting a top view of a portion **406** of Block **2** of plane **402**. As can be seen from FIG. 4B, the block depicted in FIG. 4B extends in the direction of **432**. In one embodiment, the memory array has many layers; however, FIG. 4B only shows the top layer.

FIG. 4B depicts a plurality of circles that represent the memory holes, which are also referred to as vertical columns. Each of the memory holes/vertical columns include multiple select transistors (also referred to as a select gate or selection gate) and multiple memory cells. In one embodiment, each memory hole/vertical column implements a NAND string. For example, FIG. 4B labels a subset of the memory holes/vertical columns/NAND strings **432**, **436**, **446**, **456**, **462**, **466**, **472**, **474** and **476**.

FIG. 4B also depicts a set of bit lines **415**, including bit lines **411**, **412**, **413**, **414**, . . . **419**. FIG. 4B shows twenty four

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bit lines because only a portion of the block is depicted. It is contemplated that more than twenty four bit lines connected to memory holes/vertical columns of the block. Each of the circles representing memory holes/vertical columns has an "x" to indicate its connection to one bit line. For example, bit line **411** is connected to memory holes/vertical columns **436**, **446**, **456**, **466** and **476**.

The block depicted in FIG. 4B includes a set of isolation regions **482**, **484**, **486** and **488**, which are formed of SiO₂; however, other dielectric materials can also be used. Isolation regions **482**, **484**, **486** and **488** serve to divide the top layers of the block into five regions; for example, the top layer depicted in FIG. 4B is divided into regions **430**, **440**, **450**, **460** and **470**. In one embodiment, the isolation regions only divide the layers used to implement select gates so that NAND strings in different regions can be independently selected. In one example implementation, a bit line connects to one memory hole/vertical column/NAND string in each of regions **430**, **440**, **450**, **460** and **470**. In that implementation, each block has twenty four rows of active columns and each bit line connects to five rows in each block. In one embodiment, all of the five memory holes/vertical columns/NAND strings connected to a common bit line are connected to the same set of word lines; therefore, the system uses the drain side selection lines to choose one (or another subset) of the five to be subjected to a memory operation (program, verify, read, and/or erase).

FIG. 4B also shows Line Interconnects LI, which are metal connections to the source line SL from above the memory array. Line Interconnects LI are positioned adjacent regions **430** and **470**.

Although FIG. 4B shows each region **430**, **440**, **450**, **460** and **470** having four rows of memory holes/vertical columns, five regions and twenty four rows of memory holes/vertical columns in a block, those exact numbers are an example implementation. Other embodiments may include more or less regions per block, more or less rows of memory holes/vertical columns per region and more or less rows of vertical columns per block. FIG. 4B also shows the memory holes/vertical columns being staggered. In other embodiments, different patterns of staggering can be used. In some embodiments, the memory holes/vertical columns are not staggered.

FIG. 4C depicts a portion of one embodiment of a three dimensional memory structure **202** showing a cross-sectional view along line AA of FIG. 4B. This cross sectional view cuts through memory holes/vertical columns (NAND strings) **472** and **474** of region **470** (see FIG. 4B). The structure of FIG. 4C includes three drain side select layers SGD0, SGD1 and SGD2; three source side select layers SGS0, SGS1, and SGS2; three drain side GIDL generation transistor layers SGDT0, SGDT1, and SGDT2; three source side GIDL generation transistor layers SGSB0, SGSB1, and SGSB2; four drain side dummy word line layers DD0, DD1, DD2 and DD3; four source side dummy word line layers DS0, DS1, DS2 and DS3; dummy word line layers DU and DL; one hundred and sixty two word line layers WL0-WL161 for connecting to data memory cells, and dielectric layers DL. Other embodiments can implement more or less than the numbers described above for FIG. 4C. In one embodiment, SGD0, SGD1 and SGD2 are connected together; and SGS0, SGS1 and SGS2 are connected together. In other embodiments, more or less number of SGDs (greater or lesser than three) are connected together, and more or less number of SGSs (greater or lesser than three) are connected together.

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In one embodiment, erasing the memory cells is performed using gate induced drain leakage (GIDL), which includes generating charge carriers at the GIDL generation transistors such that the carriers get injected into the charge trapping layers of the NAND strings to change threshold voltage of the memory cells. FIG. 4C shows three GIDL generation transistors at each end of the NAND string; however, in other embodiments there are more or less than three. Embodiments that use GIDL at both sides of the NAND string may have GIDL generation transistors at both sides. Embodiments that use GIDL at only the drain side of the NAND string may have GIDL generation transistors only at the drain side. Embodiments that use GIDL at only the source side of the NAND string may have GIDL generation transistors only at the source side.

FIG. 4C shows three GIDL generation transistors at each end of the NAND string. It is likely that charge carriers are only generated by GIDL at one of the three GIDL generation transistors at each end of the NAND string. Based on process variances during manufacturing, it is likely that one of the three GIDL generation transistors at an end of the NAND string is best suited for GIDL. For example, the GIDL generation transistors have an abrupt pn junction to generate the charge carriers for GIDL and, during fabrication, a phosphorous diffusion is performed at the polysilicon channel of the GIDL generation transistors. In some cases, the GIDL generation transistor with the shallowest phosphorous diffusion is the GIDL generation transistor that generates the charge carriers during erase. However, in some embodiments charge carriers can be generated by GIDL at multiple GIDL generation transistors at a particular side of the NAND string.

Memory holes/Vertical columns **472** and **474** are depicted protruding through the drain side select layers, source side select layers, dummy word line layers, GIDL generation transistor layers and word line layers. In one embodiment, each memory hole/vertical column comprises a vertical NAND string. Below the memory holes/vertical columns and the layers listed below is substrate **453**, an insulating film **454** on the substrate, and source line SL. The NAND string of memory hole/vertical column **472** has a source end at a bottom of the stack and a drain end at a top of the stack. As in agreement with FIG. 4B, FIG. 4C show vertical memory hole/column **472** connected to bit line **414** via connector **417**.

For ease of reference, drain side select layers; source side select layers, dummy word line layers, GIDL generation transistor layers and data word line layers collectively are referred to as the conductive layers. In one embodiment, the conductive layers are made from a combination of TiN and Tungsten. In other embodiments, other materials can be used to form the conductive layers, such as doped polysilicon, metal such as Tungsten, metal silicide, such as nickel silicide, tungsten silicide, aluminum silicide or the combination thereof. In some embodiments, different conductive layers can be formed from different materials. Between conductive layers are dielectric layers DL. In one embodiment, the dielectric layers are made from SiO₂. In other embodiments, other dielectric materials can be used to form the dielectric layers.

The non-volatile memory cells are formed along memory holes/vertical columns which extend through alternating conductive and dielectric layers in the stack. In one embodiment, the memory cells are arranged in NAND strings. The word line layers WL0-W161 connect to memory cells (also called data memory cells). Dummy word line layers connect to dummy memory cells. A dummy memory cell does not

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store and is not eligible to store host data (data provided from the host, such as data from a user of the host), while a data memory cell is eligible to store host data. In some embodiments, data memory cells and dummy memory cells may have a same structure. Drain side select layers SGD0, SGD1, and SGD2 are used to electrically connect and disconnect NAND strings from bit lines. Source side select layers SGS0, SGS1, and SGS2 are used to electrically connect and disconnect NAND strings from the source line SL.

FIG. 4C shows that the memory array is implemented as a two tier architecture, with the tiers separated by a Joint area. In one embodiment it is expensive and/or challenging to etch so many word line layers intermixed with dielectric layers. To ease this burden, one embodiment includes laying down a first stack of word line layers (e.g., WL0-WL80) alternating with dielectric layers, laying down the Joint area, and laying down a second stack of word line layers (e.g., WL81-WL161) alternating with dielectric layers. The Joint areas are positioned between the first stack and the second stack. In one embodiment, the Joint areas are made from the same materials as the word line layers. In other embodiments, there can be no Joint area or multiple Joint areas.

FIG. 4D depicts a portion of one embodiment of a three dimensional memory structure **202** showing a cross-sectional view along line BB of FIG. 4B. This cross sectional view cuts through memory holes/vertical columns (NAND strings) **432** and **434** of region **430** (see FIG. 4B). FIG. 4D shows the same alternating conductive and dielectric layers as FIG. 4C. FIG. 4D also shows isolation region **482**. Isolation regions **482**, **484**, **486** and **488** occupy space that would have been used for a portion of the memory holes/vertical columns/NAND strings. For example, isolation region **482** occupies space that would have been used for a portion of memory hole/vertical column **434**. More specifically, a portion (e.g., half the diameter) of vertical column **434** has been removed in layers SGDT0, SGDT1, SGDT2, SGD0, SGD1, SGD2, DD0, DD1 and DD2 to accommodate isolation region **482**. Thus, while most of the vertical column **434** is cylindrical (with a circular cross section), the portion of vertical column **434** in layers SGDT0, SGDT1, SGDT2, SGD0, SGD1, SGD2, DD0, DD1 and DD2 has a semi-circular cross section. In one embodiment, after the stack of alternating conductive and dielectric layers is formed, the stack is etched to create space for the isolation region and that space is then filled in with SiO₂.

FIG. 4E depicts a cross sectional view of region **429** of FIG. 4C that includes a portion of memory hole/vertical column **472**. In one embodiment, the memory holes/vertical columns are round; however, in other embodiments other shapes can be used. In one embodiment, memory hole/vertical column **472** includes an inner core layer **490** that is made of a dielectric, such as SiO₂. Other materials can also be used. Surrounding inner core **490** is polysilicon channel **491**. Materials other than polysilicon can also be used. Note that it is the channel **491** that connects to the bit line and the source line. Surrounding channel **491** is a tunneling dielectric **492**. In one embodiment, tunneling dielectric **492** has an ONO structure. Surrounding tunneling dielectric **492** is charge trapping layer **493**, such as (for example) Silicon Nitride. Other memory materials and structures can also be used. The technology described herein is not limited to any particular material or structure.

FIG. 4E depicts dielectric layers DL as well as word line layers WL160, WL159, WL158, WL157, and WL156. Each of the word line layers includes a word line region **496** surrounded by an aluminum oxide layer **497**, which is

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surrounded by a blocking oxide layer 498. In other embodiments, the blocking oxide layer can be a vertical layer parallel and adjacent to charge trapping layer 493. The physical interaction of the word line layers with the vertical column forms the memory cells. Thus, a memory cell, in one embodiment, comprises channel 491, tunneling dielectric 492, charge trapping layer 493, blocking oxide layer 498, aluminum oxide layer 497 and word line region 496. For example, word line layer WL160 and a portion of memory hole/vertical column 472 comprise a memory cell MC1. Word line layer WL159 and a portion of memory hole/vertical column 472 comprise a memory cell MC2. Word line layer WL158 and a portion of memory hole/vertical column 472 comprise a memory cell MC3. Word line layer WL157 and a portion of memory hole/vertical column 472 comprise a memory cell MC4. Word line layer WL156 and a portion of memory hole/vertical column 472 comprise a memory cell MC5. In other architectures, a memory cell may have a different structure; however, the memory cell would still be the storage unit.

When a memory cell is programmed, electrons are stored in a portion of the charge trapping layer 493 which is associated with (e.g. in) the memory cell. These electrons are drawn into the charge trapping layer 493 from the channel 491, through the tunneling dielectric 492, in response to an appropriate voltage on word line region 496. The threshold voltage (V_{th}) of a memory cell is increased in proportion to the amount of stored charge. In one embodiment, the programming is achieved through Fowler-Nordheim tunneling of the electrons into the charge trapping layer. During an erase operation, the electrons return to the channel or holes are injected into the charge trapping layer to recombine with electrons. In one embodiment, erasing is achieved using hole injection into the charge trapping layer via a physical mechanism such as GIDL.

FIG. 4F is a schematic diagram of a portion of the memory array 202 depicted in in FIGS. 4-4E. FIG. 4F shows physical data word lines WL0-WL161 running across the entire block. The structure of FIG. 4F corresponds to a portion 406 in Block 2 of FIG. 4A, including bit line 411. Within the block, in one embodiment, each bit line is connected to five NAND strings. Thus, FIG. 4F shows bit line connected to NAND string NS0 (which corresponds to memory hole/vertical column 436), NAND string NS1 (which corresponds to memory hole/vertical column 446), NAND string NS2 (which corresponds to vertical column 456), NAND string NS3 (which corresponds to memory hole/vertical column 466), and NAND string NS4 (which corresponds to memory hole/vertical column 476). As mentioned above, in one embodiment, SGD0, SGD1 and SGD2 are connected together to operate as a single logical select gate for each region separated by isolation regions (482, 484, 486 and 486) to form SGD-s0, SGD-s1, SGD-s2, SGD-s3, and SGD-s4. SGS0, SGS1 and SGS2 are also connected together to operate as a single logical select gate that is represented in FIG. 4F as SGS. Although the select gates SGD-s0, SGD-s1, SGD-s2, SGD-s3, and SGD-s4 are isolated from each other due to the isolation regions, the data word lines WL0-WL161 of each region are connected together. Thus, data word lines WL0-WL161 are connected to NAND strings (and memory cells) of each (or every) region (430, 440, 450, 460, 470) of a block.

The isolation regions (482, 484, 486 and 486) are used to allow for separate control of regions 430, 440, 450, 460, 470. A first region corresponds to those vertical NAND strings controlled by SGD-s0. A second region corresponds to those vertical NAND strings controlled by SGD-s1. A

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third region corresponds to those vertical NAND strings controlled by SGD-s2. A fourth region corresponds to those vertical NAND strings controlled by SGD-s3. A fifth region corresponds to those vertical NAND strings controlled by SGD-s4.

FIG. 4F only shows NAND strings connected to bit line 411. However, a full schematic of the block would show every bit line and five vertical NAND strings (that are in separate regions) connected to each bit line.

Although the example memories of FIGS. 4-4F are three dimensional memory structure that includes vertical NAND strings with charge-trapping material, other (2D and 3D) memory structures can also be used with the technology described herein.

The memory systems discussed above can be erased, programmed and read. At the end of a successful programming process, the threshold voltages of the memory cells should be within one or more distributions of threshold voltages for programmed memory cells or within a distribution of threshold voltages for erased memory cells, as appropriate. FIG. 5A is a graph of threshold voltage versus number of memory cells, and illustrates example threshold voltage distributions for the memory array when each memory cell stores one bit of data per memory cell. Memory cells that store one bit of data per memory cell data are referred to as single level cells ("SLC"). The data stored in SLC memory cells is referred to as SLC data; therefore, SLC data comprises one bit per memory cell. Data stored as one bit per memory cell is SLC data. FIG. 5A shows two threshold voltage distributions: E and P. Threshold voltage distribution E corresponds to an erased data state. Threshold voltage distribution P corresponds to a programmed data state. Memory cells that have threshold voltages in threshold voltage distribution E are, therefore, in the erased data state (e.g., they are erased). Memory cells that have threshold voltages in threshold voltage distribution P are, therefore, in the programmed data state (e.g., they are programmed). In one embodiment, erased memory cells store data "1" and programmed memory cells store data "0." FIG. 5A depicts read reference voltage V_r . By testing (e.g., performing one or more sense operations) whether the threshold voltage of a given memory cell is above or below V_r , the system can determine a memory cells is erased (state E) or programmed (state P). FIG. 5A also depicts verify reference voltage V_v . In some embodiments, when programming memory cells to data state P, the system will test whether those memory cells have a threshold voltage greater than or equal to V_v .

FIGS. 5B-D illustrate example threshold voltage distributions for the memory array when each memory cell stores multiple bit per memory cell data. Memory cells that store multiple bit per memory cell data are referred to as multi-level cells ("MLC"). The data stored in MLC memory cells is referred to as MLC data; therefore, MLC data comprises multiple bits per memory cell. Data stored as multiple bits of data per memory cell is MLC data. In the example embodiment of FIG. 5B, each memory cell stores two bits of data. Other embodiments may use other data capacities per memory cell (e.g., such as three, four, or five bits of data per memory cell).

FIG. 5B shows a first threshold voltage distribution E for erased memory cells. Three threshold voltage distributions A, B and C for programmed memory cells are also depicted. In one embodiment, the threshold voltages in the distribution E are negative and the threshold voltages in distributions A, B and C are positive. Each distinct threshold voltage distribution of FIG. 5B corresponds to predetermined values for the set of data bits. In one embodiment, each bit of data

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of the two bits of data stored in a memory cell are in different logical pages, referred to as a lower page (LP) and an upper page (UP). In other embodiments, all bits of data stored in a memory cell are in a common logical page. The specific relationship between the data programmed into the memory cell and the threshold voltage levels of the cell depends upon the data encoding scheme adopted for the cells. Table 1 provides an example encoding scheme.

TABLE 1

	E	A	B	C
LP	1	0	0	1
UP	1	1	0	0

In one embodiment, known as full sequence programming, memory cells can be programmed from the erased data state E directly to any of the programmed data states A, B or C using the process of FIG. 6 (discussed below). For example, a population of memory cells to be programmed may first be erased so that all memory cells in the population are in erased data state E. Then, a programming process is used to program memory cells directly into data states A, B, and/or C. For example, while some memory cells are being programmed from data state E to data state A, other memory cells are being programmed from data state E to data state B and/or from data state E to data state C. The arrows of FIG. 5B represent the full sequence programming. In some embodiments, data states A-C can overlap, with memory controller 120 (or control die 211) relying on error correction to identify the correct data being stored.

FIG. 5C depicts example threshold voltage distributions for memory cells where each memory cell stores three bits of data per memory cells (which is another example of MLC data). FIG. 5C shows eight threshold voltage distributions, corresponding to eight data states. The first threshold voltage distribution (data state) Er represents memory cells that are erased. The other seven threshold voltage distributions (data states) A-G represent memory cells that are programmed and, therefore, are also called programmed states. Each threshold voltage distribution (data state) corresponds to predetermined values for the set of data bits. The specific relationship between the data programmed into the memory cell and the threshold voltage levels of the cell depends upon the data encoding scheme adopted for the cells. In one embodiment, data values are assigned to the threshold voltage ranges using a Gray code assignment so that if the threshold voltage of a memory erroneously shifts to its neighboring physical state, only one bit will be affected. Table 2 provides an example of an encoding scheme for embodiments in which each bit of data of the three bits of data stored in a memory cell are in different logical pages, referred to as a lower page (LP), middle page (MP) and an upper page (UP).

TABLE 2

	Er	A	B	C	D	E	F	G
UP	1	1	1	0	0	0	0	1
MP	1	1	0	0	1	1	0	0
LP	1	0	0	0	0	1	1	1

FIG. 5C shows seven read reference voltages, VrA, VrB, VrC, VrD, VrE, VrF, and VrG for reading data from memory cells. By testing (e.g., performing sense operations) whether the threshold voltage of a given memory cell is above or

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below the seven read reference voltages, the system can determine what data state (i.e., A, B, C, D, . . .) a memory cell is in.

FIG. 5C also shows seven verify reference voltages, VvA, VvB, VvC, VvD, VvE, VvF, and VvG. In some embodiments, when programming memory cells to data state A, the system will test whether those memory cells have a threshold voltage greater than or equal to VvA. When programming memory cells to data state B, the system will test whether the memory cells have threshold voltages greater than or equal to VvB. When programming memory cells to data state C, the system will determine whether memory cells have their threshold voltage greater than or equal to VvC. When programming memory cells to data state D, the system will test whether those memory cells have a threshold voltage greater than or equal to VvD. When programming memory cells to data state E, the system will test whether those memory cells have a threshold voltage greater than or equal to VvE. When programming memory cells to data state F, the system will test whether those memory cells have a threshold voltage greater than or equal to VvF. When programming memory cells to data state G, the system will test whether those memory cells have a threshold voltage greater than or equal to VvG. FIG. 5C also shows Vev, which is an erase verify reference voltage to test whether a memory cell has been properly erased.

In an embodiment that utilizes full sequence programming, memory cells can be programmed from the erased data state Er directly to any of the programmed data states A-G using the process of FIG. 6 (discussed below). For example, a population of memory cells to be programmed may first be erased so that all memory cells in the population are in erased data state Er. Then, a programming process is used to program memory cells directly into data states A, B, C, D, E, F, and/or G. For example, while some memory cells are being programmed from data state Er to data state A, other memory cells are being programmed from data state Er to data state B and/or from data state Er to data state C, and so on. The arrows of FIG. 5C represent the full sequence programming. In some embodiments, data states A-G can overlap, with control die 211 and/or memory controller 120 relying on error correction to identify the correct data being stored. Note that in some embodiments, rather than using full sequence programming, the system can use multi-pass programming processes known in the art.

In general, during verify operations and read operations, the selected word line is connected to a voltage (one example of a reference signal), a level of which is specified for each read operation (e.g., see read compare voltages/levels VrA, VrB, VrC, VrD, VrE, VrF, and VrG, of FIG. 5C) or verify operation (e.g. see verify target voltages/levels VvA, VvB, VvC, VvD, VvE, VvF, and VvG of FIG. 5C) in order to determine whether a threshold voltage of the concerned memory cell has reached such level. After applying the word line voltage, the conduction current of the memory cell is measured to determine whether the memory cell turned on (conducted current) in response to the voltage applied to the word line. If the conduction current is measured to be greater than a certain value, then it is assumed that the memory cell turned on and the voltage applied to the word line is greater than the threshold voltage of the memory cell. If the conduction current is not measured to be greater than the certain value, then it is assumed that the memory cell did not turn on and the voltage applied to the word line is not greater than the threshold voltage of the memory cell. During a read or verify process, the unselected memory cells are provided with one or more read pass voltages (also

referred to as bypass voltages) at their control gates so that these memory cells will operate as pass gates (e.g., conducting current regardless of whether they are programmed or erased).

There are many ways to measure the conduction current of a memory cell during a read or verify operation. In one example, the conduction current of a memory cell is measured by the rate it discharges or charges a dedicated capacitor in the sense amplifier. In another example, the conduction current of the selected memory cell allows (or fails to allow) the NAND string that includes the memory cell to discharge a corresponding bit line. The voltage on the bit line is measured after a period of time to see whether it has been discharged or not. Note that the technology described herein can be used with different methods known in the art for verifying/reading. Other read and verify techniques known in the art can also be used.

FIG. 5D depicts threshold voltage distributions when each memory cell stores four bits of data, which is another example of MLC data. FIG. 5D depicts that there may be some overlap between the threshold voltage distributions (data states) S0-S15. The overlap may occur due to factors such as memory cells losing charge (and hence dropping in threshold voltage). Program disturb can unintentionally increase the threshold voltage of a memory cell. Likewise, read disturb can unintentionally increase the threshold voltage of a memory cell. Over time, the locations of the threshold voltage distributions may change. Such changes can increase the bit error rate, thereby increasing decoding time or even making decoding impossible. Changing the read reference voltages can help to mitigate such effects. Using ECC during the read process can fix errors and ambiguities. Note that in some embodiments, the threshold voltage distributions for a population of memory cells storing four bits of data per memory cell do not overlap and are separated from each other. The threshold voltage distributions of FIG. 5D will include read reference voltages and verify reference voltages, as discussed above.

When using four bits per memory cell, the memory can be programmed using the full sequence programming discussed above, or multi-pass programming processes known in the art. Each threshold voltage distribution (data state) of FIG. 5D corresponds to predetermined values for the set of data bits. The specific relationship between the data programmed into the memory cell and the threshold voltage levels of the cell depends upon the data encoding scheme adopted for the cells. Table 3 provides an example of an encoding scheme for embodiments in which each bit of data of the four bits of data stored in a memory cell are in different logical pages, referred to as a lower page (LP), middle page (MP), an upper page (UP) and top page (TP).

TABLE 3

	S0	S1	S2	S3	S4	S5	S6	S7	S8	S9	S10	S11	S12	S13	S14	S15
TP	1	1	1	1	1	0	0	0	0	0	1	1	0	0	0	1
UP	1	1	0	0	0	0	0	0	1	1	1	1	1	1	0	0
MP	1	1	1	0	0	0	0	1	1	0	0	0	0	1	1	1
LP	1	0	0	0	1	1	0	0	0	0	0	1	1	1	1	1

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FIG. 6 is a flowchart describing one embodiment of a process for programming memory cells. For purposes of this document, the term program and programming are synonymous with write and writing. In one example embodiment, the process of FIG. 6 is performed for memory array 202 using the one or more control circuits (e.g., system control

logic 260, column control circuitry 210, row control circuitry 220) discussed above. In one example embodiment, the process of FIG. 6 is performed by integrated memory assembly 207 using the one or more control circuits (e.g., system control logic 260, column control circuitry 210, row control circuitry 220) of control die 211 to program memory cells on memory die 201. The process includes multiple loops, each of which includes a program phase and a verify phase. The process of FIG. 6 is performed to implement the full sequence programming, as well as other programming schemes including multi-stage programming. When implementing multi-stage programming, the process of FIG. 6 is used to implement any/each stage of the multi-stage programming process.

Typically, the program voltage applied to the control gates (via a selected data word line) during a program operation is applied as a series of program pulses (e.g., voltage pulses). Between programming pulses are a set of verify pulses (e.g., voltage pulses) to perform verification. In many implementations, the magnitude of the program pulses is increased with each successive pulse by a predetermined step size. In step 602 of FIG. 6, the programming voltage signal (V_{pgm}) is initialized to the starting magnitude (e.g., ~12-16V or another suitable level) and a program counter PC maintained by state machine 262 is initialized at 1. In one embodiment, the group of memory cells selected to be programmed (referred to herein as the selected memory cells) are programmed concurrently and are all connected to the same word line (the selected word line). There will likely be other memory cells that are not selected for programming (unselected memory cells) that are also connected to the selected word line. That is, the selected word line will also be connected to memory cells that are supposed to be inhibited from programming. Additionally, as memory cells reach their intended target data state, they will be inhibited from further programming. Those NAND strings (e.g., unselected NAND strings) that include memory cells connected to the selected word line that are to be inhibited from programming have their channels boosted to inhibit programming. When a channel has a boosted voltage, the voltage differential between the channel and the word line is not large enough to cause programming. To assist in the boosting, in step 604 the control die will pre-charge channels of NAND strings that include memory cells connected to the selected word line that are to be inhibited from programming. In step 606, NAND strings that include memory cells connected to the selected word line that are to be inhibited from programming have their channels boosted to inhibit programming. Such NAND strings are referred to herein as "unselected NAND strings." In one embodiment, the unselected word lines receive one or more boosting voltages (e.g., ~7-11 volts) to

perform boosting schemes. A program inhibit voltage is applied to the bit lines coupled the unselected NAND string.

In step 608, a program voltage pulse of the programming voltage signal V_{pgm} is applied to the selected word line (the word line selected for programming). If a memory cell on a NAND string should be programmed, then the correspond-

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ing bit line is biased at a program enable voltage. In step 608, the program pulse is concurrently applied to all memory cells connected to the selected word line so that all of the memory cells connected to the selected word line are programmed concurrently (unless they are inhibited from programming). That is, they are programmed at the same time or during overlapping times (both of which are considered concurrent). In this manner all of the memory cells connected to the selected word line will concurrently have their threshold voltage change, unless they are inhibited from programming.

In step 610, program verify is performed and memory cells that have reached their target states are locked out from further programming by the control die. Step 610 includes performing verification of programming by sensing at one or more verify reference levels. In one embodiment, the verification process is performed by testing whether the threshold voltages of the memory cells selected for programming have reached the appropriate verify reference voltage. In step 610, a memory cell may be locked out after the memory cell has been verified (by a test of the V_t) that the memory cell has reached its target state.

In one embodiment of step 610, a smart verify technique is used such that the system only verifies a subset of data states during a program loop (steps 604-628). For example, the first program loop includes verifying for data state A (see FIG. 5C), depending on the result of the verify operation the second program loop may perform verify for data states A and B, depending on the result of the verify operation the third program loop may perform verify for data states B and C, and so on.

In step 616, the number of memory cells that have not yet reached their respective target threshold voltage distribution are counted. That is, the number of memory cells that have, so far, failed to reach their target state are counted. This counting can be done by state machine 262, memory controller 120, or another circuit. In one embodiment, there is one total count, which reflects the total number of memory cells currently being programmed that have failed the last verify step. In another embodiment, separate counts are kept for each data state.

In step 617, the system determines whether the verify operation in the latest performance of step 610 included verifying for the last data state (e.g., data state G of FIG. 5C). If so, then in step 618, it is determined whether the count from step 616 is less than or equal to a predetermined limit. In one embodiment, the predetermined limit is the number of bits that can be corrected by error correction codes (ECC) during a read process for the page of memory cells. If the number of failed cells is less than or equal to the predetermined limit, then the programming process can stop and a status of "PASS" is reported in step 614. In this situation, enough memory cells programmed correctly such that the few remaining memory cells that have not been completely programmed can be corrected using ECC during the read process. In some embodiments, the predetermined limit used in step 618 is below the number of bits that can be corrected by error correction codes (ECC) during a read process to allow for future/additional errors. When programming less than all of the memory cells for a page, the predetermined limit can be a portion (pro-rata or not pro-rata) of the number of bits that can be corrected by ECC during a read process for the page of memory cells. In some embodiments, the limit is not predetermined. Instead, it changes based on the number of errors already counted for the page, the number of program-erase cycles performed or other criteria.

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If in step 617 it was determined that the verify operation in the latest performance of step 610 did not include verifying for the last data state or in step 618 it was determined that the number of failed memory cells is not less than the predetermined limit, then in step 619 the data states that will be verified in the next performance of step 610 (in the next program loop) is adjusted as per the smart verify scheme discussed above. In step 620, the program counter PC is checked against the program limit value (PL). Examples of program limit values include 6, 12, 16, 19, 20 and 30; however, other values can be used. If the program counter PC is not less than the program limit value PL, then the program process is considered to have failed and a status of FAIL is reported in step 624. If the program counter PC is less than the program limit value PL, then the process continues at step 626 during which time the Program Counter PC is incremented by 1 and the programming voltage signal V_{pgm} is stepped up to the next magnitude. For example, the next pulse will have a magnitude greater than the previous pulse by a step size ΔV_{pgm} (e.g., a step size of 0.1-1.0 volts). After step 626, the process continues at step 604 and another program pulse is applied to the selected word line (by the control die) so that another program loop (steps 604-626) of the programming process of FIG. 6 is performed.

In one embodiment memory cells are erased prior to programming. Erasing is the process of changing the threshold voltage of one or more memory cells from a programmed data state to an erased data state. For example, changing the threshold voltage of one or more memory cells from state P to state E of FIG. 5A, from states A/B/C to state E of FIG. 5B, from states A-G to state Er of FIG. 5C or from states S1-S15 to state S0 of FIG. 5D.

One technique to erase memory cells in some memory devices is to bias a p-well (or other types of) substrate to a high voltage to charge up a NAND channel. An erase enable voltage (e.g., a low voltage) is applied to control gates of memory cells while the NAND channel is at a high voltage to erase the memory cells. Herein, this is referred to as p-well erase.

Another approach to erasing memory cells is to generate gate induced drain leakage ("GIDL") current to charge up the NAND string channel. An erase enable voltage is applied to control gates of the memory cells, while maintaining the NAND string channel potential to erase the memory cells. Herein, this is referred to as GIDL erase. Both p-well erase and GIDL erase may be used to lower the threshold voltage (V_t) of memory cells.

In one embodiment, the GIDL current is generated by causing a drain-to-gate voltage at a GIDL generation transistor (e.g., transistors connected to SGDT0, SGDT1, SGDT2, SGSB0, SGSB1 and SGSB2). In some embodiments, a select gate (e.g., SGD or SGS) can be used as a GIDL generation transistor. A transistor drain-to-gate voltage that generates a GIDL current is referred to herein as a GIDL voltage. The GIDL current may result when the GIDL generation transistor drain voltage is significantly higher than the GIDL generation transistor control gate voltage. GIDL current is a result of carrier generation, i.e., electron-hole pair generation due to band-to-band tunneling and/or trap-assisted generation. In one embodiment, GIDL current may result in one type of carriers (also referred to a charge carriers), e.g., holes, predominantly moving into the NAND channel, thereby raising or changing the potential of the channel. The other type of carriers, e.g., electrons, are extracted from the channel, in the direction of a bit line or in the direction of a source line, by an electric field. During

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erase, the holes may tunnel from the channel to a charge storage region of the memory cells (e.g., to charge trapping layer 493) and recombine with electrons there, to lower the threshold voltage of the memory cells.

The GIDL current may be generated at either end (or both ends) of the NAND string. A first GIDL voltage may be created between two terminals of a GIDL generation transistor (e.g., connected to SGDT0, SGDT1, SGDT2) that is connected to or near a bit line to generate a first GIDL current. A second GIDL voltage may be created between two terminals of a GIDL generation transistor (e.g., SGSB0, SGSB1 and SGSB2) that is connected to or near a source line to generate a second GIDL current. Erasing based on GIDL current at only one end of the NAND string is referred to as a one-sided GIDL erase. Erasing based on GIDL current at both ends of the NAND string is referred to as a two-sided GIDL erase. The technology described herein can be used with one-sided GIDL erase and two-sided GIDL erase.

FIG. 7 is a flow chart describing one embodiment of a process for reading data. The process of FIG. 7 can be performed by any one of the one or more control circuits discussed above. The process of FIG. 7 can be performed entirely by a control circuit on memory die 200 (see FIG. 2A) or entirely by a control circuit on integrated memory assembly 207 (see FIG. 2B), rather than by memory controller 120. In one example, the process of FIG. 7 is performed by or at the direction of state machine 262, using other components of System Control Logic 260, Column Control Circuitry 210 and Row Control Circuitry 220. In another embodiment, the process of FIG. 7 is performed by or at the direction of memory controller 120.

In step 702, a request to read data is received by the control circuit (e.g., from the host, the memory controller, or other entity). The request to read data includes one or more addresses indicating where the data is stored in non-volatile memory 202. In step 704, the control circuit applies voltage signals to the source line, source select line SGS and drain select line SGD. In step 706, a lower overdrive voltage, VreadL, is applied to memory cells at the ends of the NAND string (e.g., at the bottom of each tier). In step 708, the standard overdrive voltage, Vread, is applied to the unselected memory cells not at the bottom of the tiers and not adjacent the selected memory cells. In step 710, a higher overdrive voltage, VreadK, is applied to those memory cells of the NAND strings that are neighbors to the selected memory cell (e.g., by applying VreadK to the word lines adjacent to the selected word line WLn). In step 712, the read compare voltage, Vcgr, is applied to the selected memory cells (e.g., by applying Vcgr to the selected word line WLn). In step 714, the control circuit senses the current through the selected memory cell in response to the voltages applied in steps 702-714. In step 716, the sensed data is decoded (e.g., removing ECC). The decoded data is reported to the host in step 718.

In many embodiments, the system will read multiple memory cells at the same time. Therefore, the process of 704-716 will be performed concurrently for multiple NAND strings of the same block such that multiple memory cells will be read concurrently.

FIG. 8 shows three NAND strings 650, 652 and 654 of a block of memory cells implementing the process of FIG. 7. In the example, the block of NAND strings will include 162 word lines (WL0-WL161). In the depicted example, word line WLn is selected for reading; therefore, all the memory cells connected to WLn will be read. As per step 712 of FIG. 7, FIG. 8 shows WLn receiving Vcgr, the read compare

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voltage. As per step 710 of FIG. 7, FIG. 8 shows neighboring word lines WLn-1 and WLn+1 receiving the higher read pass voltage, VreadK. As per step 708 of FIG. 7, FIG. 8 shows the word lines that are not connected to end memory cells receiving the standard overdrive voltage, Vread. As per step 704 of FIG. 7, FIG. 8 shows the end word line (in the example, the word lines at the bottom of the NAND string), WL0, receiving the lower overdrive voltage VreadL. In this embodiment, the various voltages Vread, VreadL and VreadK are provided to the control gates of the respective memory cells by applying the voltages to the word lines, which are connected to the control gates of memory cells. As seen from FIG. 8, each word line is connected to the control gate of multiple memory cells so that the read and programming operations are performed in parallel.

FIG. 9 is a timing diagram depicting the behavior of various signals during a read operation (e.g., the process of FIG. 7). For example, if the memory cells are SLC memory cells, the process of FIG. 9 may be performed in parallel for each memory cell during a read or verify process. If the memory cells are multi-state memory cells, the process of FIG. 9 may be performed multiple times for each memory cell in parallel during a read or verify process. For example, when reading data from a memory cell with eight states, the read process may be performed once with Vcgr=Vra, once with Vcgr=Vrb, once with Vcgr=Vrc, once with Vcgr=Vrd, once with Vcgr=Vre, once with Vcgr=Vrf, and once with Vcgr=Vrg. Other sets of read values can also be used and some embodiments will perform less than all of the read operations based on a need to read only a subset of pages of data or other reasons.

FIG. 9 depicts the following signals: BL(sel), BL(unsel), SGD(sel), SGD(unsel), WLunsel, WLunsel_neighbor (e.g., WLn+1 and WLn-1), WLunsel_end (e.g., WL0 and WL81), WLn, SGS and SL. The signal BL(sel) is the voltage applied to bit lines connected to NAND strings having memory cells selected for sensing/reading. The signal BL(unsel) is the voltage applied to bit lines connected to NAND strings that do not have any memory cells selected for sensing/reading. In some embodiments, all bit lines will be selected for sensing/reading. The signal SGD(sel) is the voltage applied to the drain side select (SGD) lines (e.g., SGD0, SGD1 and SGD2 connected together for one region) for the region (e.g., regions 420, 430, 440, 450, 460 and 470) selected for sensing/reading. The signal SGD(unsel) is the voltage applied to the drain side select (SGD) lines (e.g., SGD0, SGD1 and SGD2 connected together for one region) for the region (e.g., regions 420, 430, 440, 450, 460 and 470) not selected for sensing/reading. The signal SGS is the voltage applied to the source side select (SGS) lines (e.g., SGS0, SGS1 and SGS2 connected together). The signal SL is the voltage applied to the source line. The signal WLn is the voltage applied to the word line selected for sensing/reading. The signal WLunsel_neighbor is the voltage applied to the word lines adjacent to the selected word line (WLn), such as WLn+1 and WLn-1, both of which are not selected for sensing/reading. The signal WLunsel_end is the voltage applied to the word lines at the bottom of the tiers, such as WL0 and WL81, both of which are not selected for sensing/reading. The signal WLunsel is the voltage applied to the word lines not selected to be sensed/read other than WLunsel_neighbor and WLunsel_end.

At time t0 of FIG. 9, all signals depicted in FIG. 13 are at Vss (ground or 0 volts). At time t1, BL(sel) is raised to Vbl (e.g., 0.5-1.5v), SGD(sel) is raised to Vsg (e.g., 3.5-6v), WLunsel is raised to Vread (e.g., 6 volts), WLunsel_neighbor is raised to VreadK (e.g., 8 volts), and WLunsel_end is

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raised to VreadL (e.g., 4 volts). Also, at t1, a voltage spike is applied to WLn and subsequent to the voltage on WLn settling down then WLn is raised to Vcgr. At time t2, SGS is raised to Vsg, which provides a path for the bit line voltage to dissipate. If Vcgr is greater than the threshold voltage of the selected memory cells, then the selected memory cells will conduct current and the bit line voltage will dissipate via the source line, as depicted by curve 904. If Vcgr is not greater than the threshold voltage of the selected memory cells, then the selected memory cells will not conduct current and the bit line voltage will not dissipate via the source line, as depicted by curve 902. The sense amplifiers will sense whether the selected memory cell conducted or not at time t3. At time t4, BL(sel) is lowered to Vss. At time t5, SGD(sel), WLunsel, WLn, WLunsel_neighbor, WLunsel_end and SGS are lowered to Vss.

FIG. 10 depicts the example threshold voltage distributions of FIG. 5C, including threshold voltage distributions 1002, 1004, 1006, 1008, 1010, 1012, 1014 and 1016. If the overdrive voltages are too high, a read disturb can occur. For example, repeatedly applying VreadK to WLn+1 and WLn-1 could result in a read disturb in the form of memory cells that are supposed to be in the erased data state experience electrons migrating into the charge trapping region, thereby raising the threshold voltage of those memory cells. This is graphically depicted in FIG. 10 by threshold voltage distribution 1002a, which partially overlaps with threshold voltage distributions 1004 for the lowest programmed data state A. The shifting of the threshold voltage distribution for the erased data state (from 1002 to 1002a) results in a reduced (including the possibility of elimination of the) margin between the threshold voltage distribution for the erased data state and the threshold voltage distribution 1004 for the lowest programmed data state A. This reduced margin (including overlap) can lead to errors when reading the data. To address this Er-A margin issue (the reduced margin between the threshold voltage distribution for the erased data state 1002a and the threshold voltage distribution 1004 for the lowest programmed data state A), it is proposed that the memory system (e.g., the control circuit) adjusts the overdrive voltage applied to word lines adjacent the selected word line in order to compensate for margin degradation between the erased data state and the lowest programmed data state.

However, it has been determined that not all word lines exhibit the reduced Er-A margin issue and not all word lines will benefit from an adjustment to the overdrive voltage; therefore, the adjusting of the overdrive voltage applied to word lines adjacent the selected word line should only be performed for word lines determined to suffer from the reduced Er-A margin and that will benefit from an adjustment to the overdrive voltage. That is, after repeatedly reading a memory without reprogramming, the memory cells on some word lines will suffer from the reduced Er-A margin while the memory cells on other word lines will not suffer from the reduced Er-A margin. Thus, it is proposed to perform experimentation and/or simulation to determine which word lines will suffer from the reduced Er-A margin and will benefit from an adjustment to the overdrive voltage. For some word lines, reducing the overdrive voltage can result in a higher FBC when reading. Therefore, experimentation and/or simulation to determine which word lines that suffer from the reduced Er-A margin can implement a reduced overdrive voltage without increasing the FBC.

FIG. 11 is a flow chart describing one embodiment of a process for reading that includes adjusting the overdrive voltage applied to word lines adjacent the selected word line

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in order to compensate for margin degradation between the erased data state and the lowest programmed data state. The process of FIG. 11 can be performed by any one of the one or more control circuits discussed above. The process of FIG. 11 can be performed entirely by a control circuit on memory die 200 (see FIG. 2A) or entirely by a control circuit on integrated memory assembly 207 (see FIG. 2B), rather than by memory controller 120. In one example, the process of FIG. 11 is performed by or at the direction of state machine 262, using other components of System Control Logic 260, Column Control Circuitry 210 and Row Control Circuitry 220. In another embodiment, the process of FIG. 11 is performed by or at the direction of memory controller 120.

In step 1102 of FIG. 11, the control circuit applies one or more read compare voltages (e.g., Vcgr) to a selected word line (e.g., WLn). In step 1104, the control circuit applies a standard overdrive voltage to unselected word lines other than the one or more word lines adjacent to the selected word line. The standard overdrive voltage has been referred to above as Vread. In one embodiment, Vread is a higher voltage magnitude than any of the threshold voltages associated with the programmed data states such that applying Vread will cause the memory cells to turn on, conduct current and act as a pass gate. As mentioned above, the process of FIG. 11 includes adjusting the overdrive voltage (VreadK) applied to word lines adjacent the selected word line. The traditional overdrive voltage applied to word lines adjacent the selected word line is referred to as the nominal neighbor word line overdrive voltage. In one embodiment, the control circuit adjusts VreadK by using a voltage that is lower than the nominal neighbor word line overdrive voltage (and this lower voltage is referred to as the lower neighbor word line overdrive voltage). For example, the nominal neighbor word line overdrive voltage can be VreadK and the lower neighbor word line overdrive voltage can be VreadK-Δ. In one example, Δ=0.4v. However, as discussed above, not all word lines exhibit the reduced margin and not all word lines that will benefit from an adjustment to the overdrive voltage; therefore, the adjusting of the overdrive voltage applied to word lines adjacent the selected word line should only be performed for word lines determined to suffer from the reduced Er-A margin and that will benefit from a change. Thus, in step 1106, the control circuit applies the nominal neighbor word line overdrive voltage (e.g., VreadK) to a word line adjacent to the selected word line if the word line adjacent to the selected word line does not exhibit an increased margin between the erased data state and the lowest programmed data state in response to using a lower neighbor word line overdrive voltage. In step 1108, the control circuit applies the lower neighbor word line overdrive voltage (e.g., VreadK-Δ) to the word line adjacent to the selected word line if the word line adjacent to the selected word line exhibits an increased margin between the erased data state and the lowest programmed data state and does not exhibit a higher failed bit count in response to using the lower neighbor word line overdrive voltage. The lower neighbor word line overdrive voltage (e.g., VreadK-Δ) is a greater voltage than the standard overdrive voltage (e.g., Vread).

In one example implementation of the process of FIG. 11, the word lines that do not exhibit an increased margin between the erased data state and the lowest programmed data state in response to using the lower neighbor word line overdrive voltage on the respective word line form a first zone of word lines and the word lines that exhibit an increased margin between the erased data state and the

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lowest programmed data state in response to using the lower neighbor word line overdrive voltage on the respective word line form a second zone of word lines.

The first zone of word lines is a non-contiguous set of the word lines, meaning that not all the word lines in the first zone are in numerical order and there can be breaks in between the word lines of the first zone. For example, the first zone of word lines could include word lines WL0-10 and WL40-60. As there is a break in the word lines (e.g., WL11-WL39 are not in the first zone), the first zone of word lines is a non-contiguous set of the word lines. Similarly, the second zone of word lines is also a non-contiguous set of the word lines. This concept of two zones of word lines, each being a non-contiguous set of the word lines, is depicted in FIG. 12A.

FIG. 12A is a graph. The vertical axis is the magnitude of the overdrive voltage applied to the word line when that word line is adjacent a selected word line. The horizontal axis is word line position. FIG. 12A shows an example where the first zone (Zone 0) includes WL0-WL40, WL81-WL115 and WL131-WL161; and the second zone (Zone 1) includes WL41-WL80 and WL116-WL129. The overdrive voltage applied to word lines of Zone 0 when those word lines are adjacent the selected word line (i.e., when those word lines are $WLn+1$ or $WLn-1$) is the nominal neighbor word line overdrive voltage (e.g., V_{readK}). The overdrive voltage applied to word lines of Zone 1 when those word lines are adjacent the selected word line (i.e., when those word lines are $WLn+1$ or $WLn-1$) is the lower neighbor word line overdrive voltage (e.g., $V_{readK}-\Delta$). FIG. 12A shows that both Zone 0 and Zone 1 include a non-contiguous set of the word lines (e.g., Zone 0 has a break between WL40 and WL81). FIG. 12A also shows that Zone 0 is interleaved with Zone 1 (e.g., the part of Zone 1 consisting of WL41-WL80 is between WL0-WL40 and WL81-WL115 of Zone 0).

FIG. 12B shows three NAND strings 650, 652 and 654 of FIG. 8 during the read process of FIG. 11, when the word lines adjacent the selected word line are in Zone 1. For example, the selected word line is WL75 and the word lines adjacent the selected word line are WL74 and WL76. As mentioned above, WL74 and WL76 are in Zone 1 so they receive $V_{readK}-\Delta$ in step 1108, while WL75 receives V_{cgr} in step 1102, and the other word lines receive V_{read} in step 1104. Thus, in step 1108 the control circuit is configured to apply the lower neighbor word line overdrive voltage to two word lines adjacent to the selected word line because it has been previously determined that the two word lines adjacent to the selected word line exhibit an increased margin between the erased data state and the lowest programmed data state and do not exhibit a higher failed bit count in response to using the lower neighbor word line overdrive voltage.

FIG. 12C shows three NAND strings 650, 652 and 654 of FIG. 8 during the read process of FIG. 11, when the word lines adjacent the selected word line are in Zone 0. For example, the selected word line is WL100 and the word lines adjacent the selected word line are WL99 and WL101. As mentioned above, WL99 and WL101 are in Zone 0 so they receive V_{readK} in step 1108, while WL100 received V_{cgr} in step 1102, and the other word lines receive V_{read} in step 1104.

In the example of FIG. 12A, there are only two zones of word lines. In other embodiments, there can be more than two zones of word lines. FIG. 12D depicts a graph for an example with four zones of word lines. In other embodiments, there can be three zones of word lines or more than

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four zones of word lines. In one example, a data structure indicates which word line is in which zone. The data structure can be in volatile memory (see DS in FIG. 1) or in non-volatile memory (see DS in FIGS. 2A and 2B). The data structure can also store the magnitude of the neighbor overdrive voltage for each zone.

In the example of FIG. 12D, Zone 0 comprises WL21-WL30, WL81-WL85, and WL91-WL110. Zone 1 comprises WL0-WL20, WL31-WL50, WL86-WL90, and WL111-WL120. Zone 2 comprises WL51-WL70, WL78-WL80 and WL121-WL161. Zone 3 comprises WL71-WL77. As can be seen, Zone 0, Zone 1 and Zone 2 are each non-contiguous set of the word lines. Additionally, the zones are interleaved with each other.

The overdrive voltage applied to word lines of Zone 0 when those word lines are adjacent the selected word line (i.e., when those word lines are $WLn+1$ or $WLn-1$) is the nominal neighbor word line overdrive voltage (e.g., V_{readK}). The overdrive voltage applied to word lines of Zone 1 when those word lines are adjacent the selected word line (i.e., when those word lines are $WLn+1$ or $WLn-1$) is the lower neighbor word line overdrive voltage (e.g., $V_{readK}-\Delta$), where $\Delta=0.4v$ (or another suitable value based on the particular design). The overdrive voltage applied to word lines of Zone 2 when those word lines are adjacent the selected word line (i.e., when those word lines are $WLn+1$ or $WLn-1$) is another lower neighbor word line overdrive voltage (e.g., $V_{readK}-2\Delta$). The overdrive voltage applied to word lines of Zone 3 when those word lines are adjacent the selected word line (i.e., when those word lines are $WLn+1$ or $WLn-1$) is another lower neighbor word line overdrive voltage (e.g., $V_{readK}-3\Delta$). Thus, the neighbor word line overdrive voltages are not continuously or consistently increasing from zone to zone in a word line direction. In one embodiment, each zone is associated with a different neighbor word line overdrive voltage configured to increase margin between the erased data state and the lowest programmed data state for word lines in the respective zone and not increase failed bit count during reading of the memory cells.

FIG. 13 is a flow chart describing another embodiment of a process for reading that includes adjusting the overdrive voltage applied to word lines adjacent the selected word line in order to compensate for margin degradation between the erased data state and the lowest programmed data state. The process of FIG. 13 can be performed by any one of the one or more control circuits discussed above. The process of FIG. 13 can be performed entirely by a control circuit on memory die 200 (see FIG. 2A) or entirely by a control circuit on integrated memory assembly 207 (see FIG. 2B), rather than by memory controller 120. In one example, the process of FIG. 13 is performed by or at the direction of state machine 262, using other components of System Control Logic 260, Column Control Circuitry 210 and Row Control Circuitry 220. In another embodiment, the process of FIG. 13 is performed by or at the direction of memory controller 120.

The process of FIG. 13 is an example implementation of the process of FIG. 11 that employs multiple zones of word lines (e.g., using the embodiment of FIG. 12D). That is, for example, the word lines are divided into a set of zones and each zone is associated with a different neighbor word line overdrive voltage. In some embodiments, multiple zones can have a same neighbor word line overdrive voltage. In step 1302, the control circuit applies one or more read compare voltages (V_{cgr}) to the selected word line (WLn). In step 1304, the control circuit applies a standard overdrive voltage

(e.g., Vread) to unselected word lines other than one or more word lines adjacent to the selected word line. In step 1306, the control circuit applies a neighbor word line overdrive voltage to a neighbor word line(s) adjacent to the selected word line that is associated with the zone for the neighbor word line. The control circuit then senses the data from the selected memory cells connected to the selected word lines based on the above-described applied voltages. In one embodiment, the neighbor word line overdrive voltage applied to the neighbor word line adjacent to the selected word line that is associated with the zone for the neighbor word line is VreadK-x4, where x is chosen based on which zone the neighbor word line is in.

One example implementation of the process of FIG. 13 includes dividing the word lines into the four zones (Zone 0, Zone 1, Zone 2 and Zone 3) of FIG. 12D. FIG. 14A shows three NAND strings 650, 652 and 654 of FIG. 8 during the read process of FIG. 13, when the word lines adjacent the selected word line are in Zone 0. For example, the selected word line is WL25 and the word lines adjacent the selected word line are WL24 and WL26. As mentioned above, WL24 and WL26 are on Zone 0 so they receive VreadK in step 1306, while WL25 receives Vcgr in step 1302 and the other word lines receive Vread in step 1304.

FIG. 14B shows three NAND strings 650, 652 and 654 of FIG. 8 during the read process of FIG. 13, when the word lines adjacent the selected word line are in Zone 1. For example, the selected word line is WL113 and the word lines adjacent the selected word line are WL112 and WL114. As mentioned above, WL112 and WL114 are in Zone 1 so they receive VreadK-Δ, in step 1306, while WL113 receives Vcgr in step 1302 and the other word lines receive Vread in step 1304. Thus, in step 1306 the control circuit is configured to apply the lower neighbor word line overdrive voltage to two word lines adjacent to the selected word line because the two word lines adjacent to the selected word line exhibit an increased margin between the erased data state and the lowest programmed data state and do not exhibit a higher failed bit count in response to using the lower neighbor word line overdrive voltage.

FIG. 14C shows three NAND strings 650, 652 and 654 of FIG. 8 during the read process of FIG. 13, when the word lines adjacent the selected word line are in Zone 2. For example, the selected word line is WL140 and the word lines adjacent the selected word line are WL139 and WL141. As mentioned above, WL139 and WL141 are in Zone 2 so they receive VreadK-2Δ in step 1306, while WL140 receives Vcgr in step 1302 and the other word lines receive Vread in step 1304. Thus, in step 1306 the control circuit is configured to apply the lower neighbor word line overdrive voltage to two word lines adjacent to the selected word line because the two word lines adjacent to the selected word line exhibit an increased margin between the erased data state and the lowest programmed data state and do not exhibit a higher failed bit count in response to using the lower neighbor word line overdrive voltage.

FIG. 14D shows three NAND strings 650, 652 and 654 of FIG. 8 during the read process of FIG. 13, when the word lines adjacent the selected word line are in Zone 3. For example, the selected word line is WL72 and the word lines adjacent the selected word line are WL71 and WL73. As mentioned above, WL71 and WL73 are on Zone 2 so they receive VreadK-3Δ in step 1306, while WL72 receives Vcgr in step 1302 and the other word lines receive Vread in step 1304. Thus, in step 1306 the control circuit is configured to apply the lower neighbor word line overdrive voltage to two word lines adjacent to the selected word line because the two

word lines adjacent to the selected word line exhibit an increased margin between the erased data state and the lowest programmed data state and do not exhibit a higher failed bit count in response to using the lower neighbor word line overdrive voltage.

As discussed above, experimentation (e.g., on actual silicon or using simulation) can be used to identify which word lines will have an increased margin between the erased state and lowest data state, and not have a higher FBC, from lowering the neighbor overdrive voltage, and then the memory can be operated with the lower neighbor overdrive voltage for those identified word lines only. FIG. 15 is a flow chart describing one embodiment of a process for configuring a non-volatile memory to adjust the overdrive voltage applied to word lines adjacent the selected word line in order to compensate for margin degradation between the erased data state and the lowest programmed data state when the word lines will have an increased margin between the erased state and lowest data state, and not have a higher FBC, from lowering the neighbor overdrive voltage. The process of FIG. 15 can be performed by the control circuit, a test circuit, manually in a lab or simulation.

Step 1502 of FIG. 15 includes setting a neighbor word line overdrive voltage to an initial magnitude (e.g., the nominal neighbor word line overdrive voltage). Step 1504 includes programming the memory cells. Step 1506 includes, for each word line of the plurality of word lines, sensing threshold voltage values of the memory cells. Step 1508 includes repeatedly reading the memory cells without intervening programming. After the repeatedly reading and for each word line of the plurality of word lines in step 1508, step 1510 includes sensing threshold voltage values of the memory cells including applying read compare voltages to selected word lines, applying the neighbor word line overdrive voltage to one or more neighbors word lines of the selected word line and applying a standard overdrive voltage to unselected word lines other than neighbor word lines. Step 1512 includes determining a margin between an erased data state and a lowest programmed data state for each word line of the plurality of word lines. Step 1514 includes, for each word line, determining a number of errors in the plurality of memory cells connected to the respective word line. Step 1516 includes repeating steps 1504-1514 using different voltage magnitudes for the neighbor word line overdrive voltage. One of the different voltage magnitudes and the initial magnitude is the nominal voltage magnitude for the neighbor word line overdrive voltage. Step 1518 includes comparing margin between the erased data state and the lowest programmed data state for each word line of the plurality of word lines for the different voltage magnitudes of the neighbor word line overdrive voltage. Step 1520 includes, for word lines that exhibited a smaller margin between the erased data state and the lowest programmed data state for a lower voltage magnitude for the neighbor word line overdrive voltage than the nominal magnitude (and that did not exhibit a higher number of errors in the plurality of memory cells connected to the respective word line when using the lower voltage magnitude for the neighbor word line overdrive voltage as compared to using the nominal voltage magnitude for the neighbor word line overdrive voltage), configuring the non-volatile memory to use the lower voltage magnitude for the neighbor word line overdrive voltage for the respective word line. Step 1522 includes, for word lines that did not exhibit a smaller margin between the erased data state and the lowest programmed data state for a lower voltage magnitudes for the neighbor word line overdrive voltage than the nominal magnitude,

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configuring the non-volatile memory to use the nominal voltage magnitude for the neighbor word line overdrive voltage for the respective word line.

FIG. 16 is a flow chart describing one embodiment of a process for configuring a non-volatile memory to adjust the overdrive voltage applied to word lines adjacent the selected word line in order to compensate for margin degradation between the erased data state and the lowest programmed data state. Step 1602 includes performing Er-A margin stress (e.g., the process of FIG. 15) for a non-volatile memory in a low used condition. A low used condition refers to a memory that is fresh (never been programmed) or has experienced only a few program/erase cycles. Step 1604 includes performing Er-A margin stress (e.g., the process of FIG. 15) for the non-volatile memory in a high used condition. A high used condition refers to a memory that has experienced a few thousand program/erase cycles (e.g., 10K program/erase cycles). Step 1606 includes combining the results of steps 1602 and 1604 (e.g., averaging).

FIG. 17 is a flow chart describing one embodiment of a process for configuring a non-volatile memory to adjust the overdrive voltage applied to word lines adjacent the selected word line in order to compensate for margin degradation between the erased data state and the lowest programmed data state. Step 1702 includes performing Er-A margin stress (e.g., the process of FIG. 15) for multiple non-volatile memories in a low used condition. Step 1704 includes performing Er-A margin stress (e.g., the process of FIG. 15) for multiple non-volatile memories in a high used condition. Step 1706 includes combining the results of steps 1702 and 1704 (e.g., averaging).

A memory system has been described such that, during a read operation for memory cells connected a selected word line, the memory system adjusts the overdrive voltage applied to word lines adjacent the selected word line in order to compensate for margin degradation between the erased data state and the lowest programmed data state.

One embodiment includes a non-volatile storage apparatus comprising a plurality of non-volatile memory cells configured to store data in a set of data states including an erased data state and a lowest programmed data state; word lines connected to the memory cells; and a control circuit connected to the word lines and the memory cells. The control circuit is configured to read the memory cells by: applying one or more read compare voltages to a selected word line; applying a standard overdrive voltage to unselected word lines other than the one or more word lines adjacent to the selected word line; applying a nominal neighbor word line overdrive voltage to a word line adjacent to the selected word line if the word line adjacent to the selected word line does not exhibit an increased margin between the erased data state and the lowest programmed data state in response to using a lower neighbor word line overdrive voltage, the lower neighbor word line overdrive voltage is a lower voltage than the nominal neighbor word line overdrive voltage; and applying the lower neighbor word line overdrive voltage to the word line adjacent to the selected word line if the word line adjacent to the selected word line exhibits an increased margin between the erased data state and the lowest programmed data state and does not exhibit a higher failed bit count in response to using the lower neighbor word line overdrive voltage, the lower neighbor word line overdrive voltage is a greater voltage than the standard overdrive voltage.

In one example implementation, the word lines that do not exhibit an increased margin between the erased data state and the lowest programmed data state in response to using

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the lower neighbor word line overdrive voltage on the respective word line form a first zone of word lines. The first zone of word lines is a non-contiguous set of the word lines. The word lines that exhibit an increased margin between the erased data state and the lowest programmed data state in response to using the lower neighbor word line overdrive voltage on the respective word line form a second zone of word lines. The second zone of word lines is a non-contiguous set of the word lines. In one example, a data structure indicates which word line is in which zone. The data structure can be in volatile memory (see DS in FIG. 1) or in non-volatile memory (see DS in FIGS. 2A and 2B).

One embodiment includes a non-volatile storage apparatus comprising a plurality of non-volatile memory cells configured to store data in a set of data states including an erased data state and a lowest programmed data state; word lines connected to the memory cells, the word lines are divided into a set zones, one or more of the zones includes a non-contiguous subset of the word lines, each zone is associated with a different neighbor word line overdrive voltage; and a control circuit connected to the word lines and the memory cells. The control circuit is configured to read the memory cells by: applying one or more read compare voltages to a selected word line, applying a standard overdrive voltage to unselected word lines other than one or more word lines adjacent to the selected word line, and applying a neighbor word line overdrive voltage to a neighbor word line adjacent to the selected word line that is associated with the zone for the neighbor word line.

One embodiment includes a method for configuring a non-volatile memory that includes a plurality of memory cells connected to a plurality of word lines. The method comprises: (a) setting a neighbor word line overdrive voltage to an initial magnitude; (b) programming the memory cells; (c) repeatedly reading the memory cells without intervening programming; (d) after the repeatedly reading and for each word line of the plurality of word lines, sensing threshold voltage values of the memory cells including applying read compare voltages to selected word lines, applying the neighbor word line overdrive voltage to one or more neighbors word lines of the selected word line and applying a standard overdrive voltage to unselected word lines other than neighbor word lines; (e) determining a margin between an erased data state and a lowest programmed data state for each word line of the plurality of word lines; (f) repeating steps (c)-(e) using different voltage magnitudes for the neighbor word line overdrive voltage, one of the different voltage magnitudes and the initial magnitude representing a nominal voltage magnitude; (g) comparing margin between the erased data state and the lowest programmed data state for each word line of the plurality of word lines for the different voltage magnitudes of the neighbor word line overdrive voltage; (h) for word lines that exhibited a smaller margin between the erased data state and the lowest programmed data state for a lower voltage magnitudes for the neighbor word line overdrive voltage than the nominal magnitude, configuring the non-volatile memory to use the lower voltage magnitude for the neighbor word line overdrive voltage for the respective word line; and (i) for word lines that did not exhibit a smaller margin between the erased data state and the lowest programmed data state for a lower voltage magnitudes for the neighbor word line overdrive voltage than the nominal magnitude, configuring the non-volatile memory to use the nominal voltage magnitude for the neighbor word line overdrive voltage for the respective word line.

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One embodiment includes a non-volatile storage apparatus comprising a plurality of non-volatile memory cells configured to store data in a set of data states including an erased data state and a lowest programmed data state; word lines connected to the memory cells, the word lines are divided into a set of zones, one or more of the zones includes a non-contiguous subset of the word lines, each zone is associated with a different neighbor word line overdrive voltage; and a control circuit connected to the word lines and the memory cells. The control circuit comprises means for, during a read operation for memory cells connected to a selected word line, adjusting the overdrive voltage applied to word lines adjacent to the selected word line in order to compensate for margin degradation between the erased data state and the lowest programmed data state. Examples of the means for adjusting includes any one of or any combination of memory controller 120, state machine 262, all or a portion of system control logic 260, all or a portion of row control circuitry 220, all or a portion of column control circuitry 210, a microcontroller, a microprocessor, and/or other similar functioned circuits (including hardware only or a combination of hardware and software/firmware) performing the processes of FIGS. 11 and/or 13.

For purposes of this document, reference in the specification to “an embodiment,” “one embodiment,” “some embodiments,” or “another embodiment” may be used to describe different embodiments or the same embodiment.

For purposes of this document, a connection may be a direct connection or an indirect connection (e.g., via one or more other parts). In some cases, when an element is referred to as being connected or coupled to another element, the element may be directly connected to the other element or indirectly connected to the other element via one or more intervening elements. When an element is referred to as being directly connected to another element, then there are no intervening elements between the element and the other element. Two devices are “in communication” if they are directly or indirectly connected so that they can communicate electronic signals between them.

For purposes of this document, the term “based on” may be read as “based at least in part on.”

For purposes of this document, without additional context, use of numerical terms such as a “first” object, a “second” object, and a “third” object may not imply an ordering of objects, but may instead be used for identification purposes to identify different objects.

For purposes of this document, the term “set” of objects may refer to a “set” of one or more of the objects.

The foregoing detailed description has been presented for purposes of illustration and description. It is not intended to be exhaustive or to limit to the precise form disclosed. Many modifications and variations are possible in light of the above teaching. The described embodiments were chosen in order to best explain the principles of the proposed technology and its practical application, to thereby enable others skilled in the art to best utilize it in various embodiments and with various modifications as are suited to the particular use contemplated. It is intended that the scope be defined by the claims appended hereto.

What is claimed is:

1. A non-volatile storage apparatus, comprising:
 - a plurality of non-volatile memory cells configured to store data in a set of data states including an erased data state and a lowest programmed data state;

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word lines connected to the memory cells; and
a control circuit connected to the word lines and the memory cells, the control circuit is configured to read the memory cells by:

- applying one or more read compare voltages to a selected word line;
 - applying a standard overdrive voltage to unselected word lines other than the one or more word lines adjacent to the selected word line;
 - applying a nominal neighbor word line overdrive voltage to a word line adjacent to the selected word line if the word line adjacent to the selected word line does not exhibit an increased margin between the erased data state and the lowest programmed data state in response to using a lower neighbor word line overdrive voltage, the lower neighbor word line overdrive voltage is a lower voltage than the nominal neighbor word line overdrive voltage; and
 - applying the lower neighbor word line overdrive voltage to the word line adjacent to the selected word line if the word line adjacent to the selected word line exhibits an increased margin between the erased data state and the lowest programmed data state and does not exhibit a higher failed bit count in response to using the lower neighbor word line overdrive voltage, the lower neighbor word line overdrive voltage is a greater voltage than the standard overdrive voltage.
2. The non-volatile storage apparatus of claim 1, wherein:
 - the word lines that do not exhibit an increased margin between the erased data state and the lowest programmed data state in response to using the lower neighbor word line overdrive voltage on the respective word line form a first zone of word lines, the first zone of word lines is a non-contiguous set of the word lines; and
 - the word lines that exhibit an increased margin between the erased data state and the lowest programmed data state in response to using the lower neighbor word line overdrive voltage on the respective word line form a second zone of word lines, the second zone of word lines is a non-contiguous set of the word lines.
 3. The non-volatile storage apparatus of claim 2, wherein: the first zone is interleaved with the second zone.
 4. The non-volatile storage apparatus of claim 2, further comprising:
 - a data structure that indicates which word line is in which zone.
 5. The non-volatile storage apparatus of claim 1, wherein:
 - the word lines are divided into a set of zones, each zone is associated with a different neighbor word line overdrive voltage;
 - a first zone of the set of zones is associated with the nominal neighbor word line overdrive voltage;
 - a second zone of the set of zones is associated with the lower neighbor word line overdrive voltage;
 - other zones of the set of zones are associated with other neighbor word line overdrive voltages; and
 - the control circuit is configured to apply a neighbor word line overdrive voltage, when reading memory cells, to a neighbor word line adjacent to the selected word line that is associated with the zone for the respective neighbor word line.
 6. The non-volatile storage apparatus of claim 5, wherein: one or more of the zones includes a non-contiguous subset of the word lines.

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7. The non-volatile storage apparatus of claim 5, wherein: multiple zones of the set of zones include a non-contiguous subset of the word lines.
8. The non-volatile storage apparatus of claim 7, wherein: the multiple zones are interleaved.
9. The non-volatile storage apparatus of claim 5, wherein: the neighbor word line overdrive voltages are not continuously increasing from zone to zone in a word line direction.
10. The non-volatile storage apparatus of claim 1, wherein: the control circuit is configured to apply the lower neighbor word line overdrive voltage to two word lines adjacent to the selected word line if the two word lines adjacent to the selected word line exhibit an increased margin between the erased data state and the lowest programmed data state and do not exhibit a higher failed bit count in response to using the lower neighbor word line overdrive voltage.
11. The non-volatile storage apparatus of claim 1, wherein: the plurality of non-volatile memory cells are arranged as a plurality of NAND strings in a three dimensional memory structure; and each NAND string includes a channel, a charge trapping layer and control gates connected to the word lines.
12. A non-volatile storage apparatus, comprising: a plurality of non-volatile memory cells configured to store data in a set of data states including an erased data state and a lowest programmed data state; word lines connected to the memory cells, the word lines are divided into a set of zones, one or more of the zones includes a non-contiguous subset of the word lines, each zone is associated with a different neighbor word line overdrive voltage; and a control circuit connected to the word lines and the memory cells, the control circuit is configured to read the memory cells by: applying one or more read compare voltages to a selected word line, applying a standard overdrive voltage to unselected word lines other than one or more word lines adjacent to the selected word line, and applying a neighbor word line overdrive voltage to a neighbor word line adjacent to the selected word line that is associated with the zone for the neighbor word line.
13. The non-volatile storage apparatus of claim 12, wherein: multiple zones of the set of zones include a non-contiguous subset of the word lines; and the multiple zones are interleaved.
14. The non-volatile storage apparatus of claim 12, wherein: each zone is associated with a different neighbor word line overdrive voltage configured to increase margin between the erased data state and the lowest programmed data state for word lines in the respective zone and not increase failed bit count during reading of the memory cells.
15. A method for configuring a non-volatile memory that includes a plurality of memory cells connected to a plurality of word lines, comprising:
- (a) setting a neighbor word line overdrive voltage to an initial magnitude;
 - (b) programming the memory cells;

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- (c) repeatedly reading the memory cells without intervening programming;
 - (d) after the repeatedly reading and for each word line of the plurality of word lines, sensing threshold voltage values of the memory cells including applying read compare voltages to selected word lines, applying the neighbor word line overdrive voltage to one or more neighbors word lines of the selected word line and applying a standard overdrive voltage to unselected word lines other than neighbor word lines;
 - (e) determining a margin between an erased data state and a lowest programmed data state for each word line of the plurality of word lines;
 - (f) repeating steps (c)-(e) using different voltage magnitudes for the neighbor word line overdrive voltage, one of the different voltage magnitudes and the initial magnitude representing a nominal voltage magnitude;
 - (g) comparing margin between the erased data state and the lowest programmed data state for each word line of the plurality of word lines for the different voltage magnitudes of the neighbor word line overdrive voltage;
 - (h) for word lines that exhibited a smaller margin between the erased data state and the lowest programmed data state for a lower voltage magnitudes for the neighbor word line overdrive voltage than the nominal magnitude, configuring the non-volatile memory to use the lower voltage magnitude for the neighbor word line overdrive voltage for the respective word line; and
 - (i) for word lines that did not exhibit a smaller margin between the erased data state and the lowest programmed data state for a lower voltage magnitudes for the neighbor word line overdrive voltage than the nominal magnitude, configuring the non-volatile memory to use the nominal voltage magnitude for the neighbor word line overdrive voltage for the respective word line.
16. The method of claim 15, further comprising: prior to the repeatedly reading and for each word line of the plurality of word lines, sensing threshold voltage values of the memory cells.
17. The method of claim 15, wherein: the plurality of memory cells are arranged as a block of memory cells; and the repeatedly reading the memory cells without intervening programming comprises repeatedly reading all memory cells of the block thousands of times without intervening programming.
18. The method of claim 15, further comprising: for each word line and for the different voltage magnitudes for the neighbor word line overdrive voltage, determining a number of errors in the plurality of memory cells connected to the respective word line; and the configuring the non-volatile memory to use the lower voltage magnitude for the neighbor word line overdrive voltage is performed for word lines that did not exhibit a higher number of errors in the plurality of memory cells connected to the respective word line when using the lower voltage magnitude for the neighbor word line overdrive voltage as compared to using the nominal voltage magnitude for the neighbor word line overdrive voltage.
19. The method of claim 15, further comprising: performing steps (a)-(g) for the non-volatile memory in a low used condition;

performing steps (a)-(g) for the non-volatile memory in a high used condition; and
combining results of steps (a)-(g) for the non-volatile memory in the low used condition and the non-volatile memory in the high used condition. 5

20. The method of claim **15**, further comprising:

performing steps (a)-(g) for multiple non-volatile memories in a low used condition;

performing steps (a)-(g) for multiple non-volatile memories in a high used condition; and 10

combining results of steps (a)-(g) for multiple non-volatile memories in the low condition state and multiple non-volatile memories in the high used condition.

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